



RISC-V Trace Control Interface Specification

RISC-V N-Trace Task Group

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Preamble



This document is in the [Ratified state](#)

No changes are allowed. Any desired or needed changes can be the subject of a follow-on new extension. Ratified extensions are never revised.

Change Log

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Chapter 1. Introduction

This document presents a standardized control interface for RISC-V trace infrastructure (such as trace encoders, trace funnels, trace sinks, ...) for the *Efficient Trace for RISC-V Version 2.0 Specification* and for the *RISC-V N-Trace (Nexus-based Trace) Specification Version 1.0.0*. Standardized control interface allows trace control software development tools to be used interchangeably with any RISC-V device implementing processor and/or data trace.

Instruction Trace is a system that collects a history of processor execution, along with other events. The trace system may be set up and controlled using a register-based interface. Hart execution activity appears on the Ingress Port and feeds into a Trace Encoder where it is compressed and formatted into trace messages. The Trace Encoder transmits trace messages to a Trace Sink. In multi-core systems, each hart has its own Trace Encoder, and typically all will connect to a Trace Funnel that aggregates the trace data from multiple sources and sends the data to a single destination.

This specification does not define the hardware interconnection between the hart and Trace Encoder, as this is defined in the *Efficient Trace for RISC-V Specification Version 2.0*. This document also does not define the hardware interconnection between the Trace Encoder and Trace Funnel, or between the Trace Encoder/Funnel and Trace Sink.

This specification allows a wide range of implementations including low-gate-count minimal instruction trace and systems with only instrumentation trace. Implementation choices include whether to support instruction trace, data trace, instrumentation trace, timestamps, external triggers, various trace sink types, and various optimization tradeoffs between gate count, features, and bandwidth requirements.

1.1. Glossary

Trace Encoder (TE for short) - Hardware module that accepts execution information from a hart and generates a stream of trace messages/packets.

Trace Message/Packet - Depending on protocol different names can be used, but it means the same. It is considered as a continuous sequence of (usually bytes) describing program and/or data flow and other events.

Trace Funnel - Hardware module that combines trace streams from multiple trace sources (Trace Encoders and/or other Trace Funnels) into a single output stream of trace messages/packets.

Trace Sink - Hardware module that accepts a stream of trace messages/packets and records them into the memory or forwards them onward in some format.

Trace Decoder - Software program that takes a recorded trace (from a Trace Sink) and produces a readable execution history.

RO - Denotes read-only bit/field - it does not mean it will return the same value each time when read.

RW - Denotes read-write bit/field - value being read may not be the same as what was written as some fields may change their values because of other reasons.

RW1C - Denotes bit/field, which can be read but you must write 1 to clear it (writing 0 will be ignored). It is used for sticky status bits to assure that these are cleared by deliberate action (write 1).

WARL - Denotes Write any, read legal bit/field/register. If a non-legal value is written, the written value is converted to a value that is supported. That value should deterministically depend on the

illegal written value and the architectural state of the trace sub-system.

W1 - Denotes write-only bit, which performs an action when 1 is written to it.

SD - Reset value of a field/register is system dependent - these fields should always have the same values at trace component reset. In many cases this may be the only value supported.

Undef - This field/register may not reset. Trace tool must write correct value before enabling the trace component.

ATB - Advanced Trace Bus, a protocol described in ARM document *AMBA ATB Protocol Specification*. This is one of alternative methods to send the trace (in addition to native Trace Sinks defined in this specification).

PIB - Pin Interface Block, a parallel or serial off-chip trace port feeding into a trace probe.

?? - Used in names refer to identical fields/registers in different components. For example **tr??Active** may mean **trTeActive** or **trTsActive**.

Chapter 2. Trace Protocols and Trace Control

There are two standard RISC-V trace protocols which will utilize this **RISC-V Trace Control Interface**:

- **RISC-V N-Trace (Nexus-based Trace) Specification**
 - Version 1.0 to be ratified together with this specification.
- **Efficient Trace for RISC-V Specification**
 - Version 2.0 (ratified May 5-th 2022).

This specification together with details provided in any of above documents should be considered as a complete guideline for any standard RISC-V trace implementation.

Trace is controlled by set of 32-bit memory-mapped registers.

Not all trace protocols and components must support all registers, bits, fields and options. This document includes a chapter [Minimal Implementation](#) which describes the smallest possible set of registers and fields, but each message protocol supported by this standard must clarify the exact meaning of supported registers/fields and bits as some of them define.

Chapter 3. Trace System Overview

This section briefly describes features of the Trace Encoder and other trace components as background for understanding some of the control interface register fields.

3.1. Trace Encoder

By monitoring the Ingress Port, the Trace Encoder determines when a program flow discontinuity has occurred and whether the discontinuity is inferable or non-inferable. An inferable discontinuity is one for which the Trace Decoder can statically determine the destination, such as a direct branch instruction in which the destination or offset is included in the opcode. Non-inferable discontinuities include all other types such as interrupts, exceptions, and indirect jump instructions.

3.1.1. Branch Trace Messaging

Branch Trace Messaging is the simplest, baseline form of instruction trace. Each program counter discontinuity results in one trace message, either a Direct or Indirect Branch Message. Linear instructions (or sequences of linear instructions) do not directly generate any trace messages/packets but overflow of counters (or exceptions) may generate corresponding packets/messages - these messages are infrequent and will not affect trace compression.

Indirect Branch Messages normally contain a compressed address to reduce bandwidth. The Trace Encoder emits a Branch With Sync Message containing the complete instruction address under certain conditions. This message type is a variant of the Direct or Indirect Branch Message and includes a full address and a field indicating the reason for the Sync.

3.1.2. Branch History Messaging

Both the Efficient Trace for RISC-V (E-Trace) Specification and the RISC-V N-Trace (Nexus-based Trace) specification define systems of messages intended to improve compression by reporting only whether conditional branches are taken by encoding each branch outcome in a single taken/not-taken bit. The destinations of non-inferable jumps and calls are reported as compressed addresses. Much better optimized compression can be achieved, but an encoder implementation will typically require more hardware.

3.1.3. Other Optimizations

Several other optimizations are possible to improve trace compression. These are optional for any Trace Encoder and there should be a way to disable optimizations in case the trace system is used with code that does not follow recommended API rules. Examples of optimizations are a Return-address stack, Branch repetition, Statically inferable jump, and Branch prediction.

3.2. Trace Sinks

The Trace Encoder transmits completed messages to a Trace Sink. This specification defines a number of different sink types, all optional, and allows an implementation to define other sink types. A Trace Encoder must have at least one sink or funnel attached to it.



Trace messages/packets are sequences of bytes. In case of wider sink width, some padding/idle bytes (or additional formatting) may be added by the sink. N-Trace format allows any number of idle bytes between messages.

3.2.1. SRAM Sink

The Trace Encoder packs trace messages into fixed-width trace words (usually bytes). These are then stored in a dedicated RAM, typically located on-chip, in a circular-buffer fashion. When the RAM has filled, it may optionally be stopped, or it may wrap and overwrite earlier trace data.

3.2.2. System Memory Sink

The Trace Encoder packs trace messages into fixed-width trace words (usually bytes). These are then stored in a range of system memory reserved for trace using a DMA-type bus controller in a circular-buffer fashion. When the memory range has been filled, it may optionally be stopped, or it may wrap and overwrite earlier trace data. This type of sink may also be used to transmit trace off-chip through, for example, a PCIe or USB port.

3.2.3. PIB Sink

The Trace Encoder sends trace messages to the PIB Sink. Each message is transmitted off-chip (as sequence of bytes) using a specific protocol described later.

3.3. ATB Bridge

The ATB Bridge allows sending RISC-V trace to Arm CoreSight infrastructure (instead of RISC-V compliant sink defined in this document) as an ATB initiator. ATB Bridge is not needed for RISC-V only systems.

ATB width is byte aligned (8, 16, 32, 64, 128) which allows transport of trace messages/packets defined as sequence of bytes.

3.4. Trace Funnel

The Trace Encoder may send trace messages to a Trace Funnel. The Funnel aggregates the trace from each of its inputs (either RISC-V Trace Encoder or another Trace Funnel) and sends the combined trace stream to its designated Trace Sink or ATB Bridge, which is one or more of the sink types above.



It is assumed that each input to the funnel (Trace Encoder or another Trace Funnel) has a unique message source ID defined (`trTeSrcID` field in the `trTeControl` register).

Chapter 4. Trace Control Interface Overview

The Trace Control interface consists of a set of 32-bit registers. The control interface can be used to set up and control a trace session, retrieve collected trace, and control any trace system components.

4.1. Trace Components

Each Trace Component is controlled by a set of 32-bit registers occupying up to 4KB of an address space. Base address of each trace component must be aligned on the 4KB boundary.

Each hart being traced must have its own separate Trace Encoder control component. A system with multiple harts must allow generating messages with a field indicating which hart is responsible for that message.

This specification defines the following trace components (*N* in at the end of symbol name denotes 0-based index of the component)

Table 1. Trace Components

Component Name	Component Type (value=symbol)	Base Address (symbol # <i>N</i>)
Trace Encoder	0x1=TRCOMP_ENCODER	trBaseEncoder <i>N</i>
Trace Funnel	0x8=TRCOMP_FUNNEL	trBaseFunnel <i>N</i>
Trace RAM Sink	0x9=TRCOMP_RAMSink	trBaseRamSink <i>N</i>
Trace PIB Sink	0xA=TRCOMP_PIBSink	trBasePibSink <i>N</i>
Trace ATB Bridge	0xE=TRCOMP_ATBBRIDGE	trBaseAtbBridge <i>N</i>



This specification does NOT address the discovery of base addresses of trace components. These base addresses (symbols in above table) must be specified as part of trace tool configuration. Connections between different trace components must be also defined. Future versions of this specification may allow a single base address to be sufficient to access and discover all trace components in the system.

4.1.1. Connections Between Components

Different components must be connected via internal busses and/or FIFO buffers. This specification does not define this interconnect logic, but the following rules must be followed:

- Each component sending a trace message/packet must assure the entire packet can be accepted by the destination component (or pushed into the FIFO buffer).
 - Sending a partial packet is NEVER allowed as it will not be possible to process and decode such a trace.
- If a component cannot send an entire message/packet it must wait until it is possible to do so.
- Tracing is typically required to be non-intrusive, and if the Trace Encoder cannot keep up with the hart it should drop the packet and wait for the receiver to be ready.
 - Once trace is allowed to resume it must issue an instruction trace synchronization message/packet so the decoder will be aware that some (unknown) amount of trace has been lost.
 - It is advisable to drain the trace pipeline to some hysteresis level before resuming - otherwise a lot of short chunks of trace may be produced.
- Optionally the Trace Encoder may be configured to stall the hart to avoid trace data loss.

- To prevent trace overflows the following techniques can be used:
 - Add a FIFO capable of holding several trace messages/packets to mitigate bursts of trace data.
 - Use wider internal busses to provide more bandwidth.
 - Make sure funnels and sinks provide the same or more bandwidth than encoders.
 - Use triggers to create trace windows/ranges to limit amount of trace data - especially in multi-core configurations.

Table 2. Allowed Connections Between Components

Input	Output	Description
Ingress Port	Trace Encoder	Ingress Port (from hart) providing raw trace to be encoded
Trace Encoder	Trace RAM Sink	Single hart tracing to RAM buffer
Trace Encoder	Trace PIB Sink	Single hart tracing via pins
Trace Encoder	Trace ATB Bridge	Single hart tracing to Arm ATB infrastructure
Trace Encoder	Trace Funnel	Sending trace from single hart to Trace Funnel (to be combined from other RISC-V trace)
Trace Funnel	Trace Funnel	Sending combined trace from multiple harts to higher level Trace Funnel (to be combined from other RISC-V trace)
Trace Funnel	Trace RAM Sink	Sending combined trace from multiple harts to RAM buffer
Trace Funnel	Trace PIB Sink	Sending combined trace from multiple harts via pins
Trace Funnel	Trace ATB Bridge	Sending combined trace from multiple harts to Arm ATB infrastructure
Trace ATB Bridge	Arm ATB bus	Sending trace to ATB (to combine RISC-V trace with other Arm components on the system)



Sending RISC-V trace to Arm CoreSight infrastructure is allowed (via ATB Bridge), but this specification does not specify how to transport trace data from other Arm CoreSight components in the system using RISC-V Trace sub-system. One of possible ways of doing so would be to create a custom trace component, configure it to encapsulate it as custom N-Trace trace messages and connect it as input to one of trace funnels.

4.1.2. Example Component Connection Diagrams

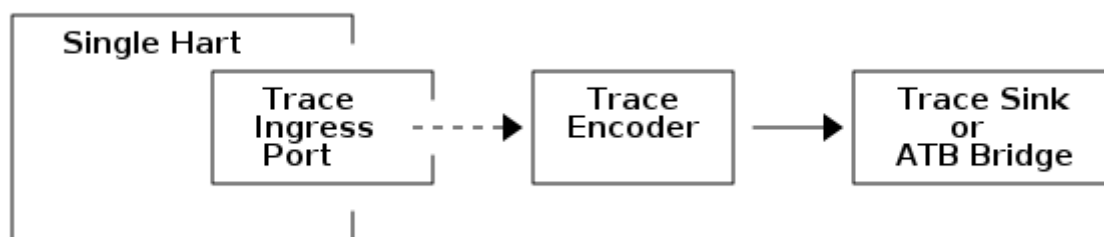


Figure 1. Simplest trace: Single Hart, Trace Encoder and Trace Sink/Bridge

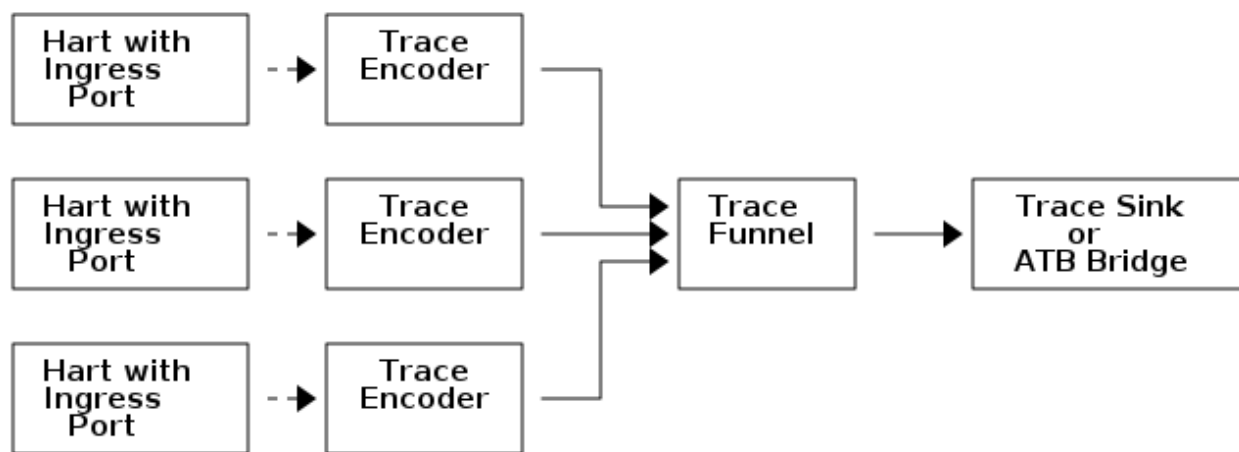


Figure 2. Multi-hart trace: Three harts, three Encoders, single Funnel and single Sink/Bridge

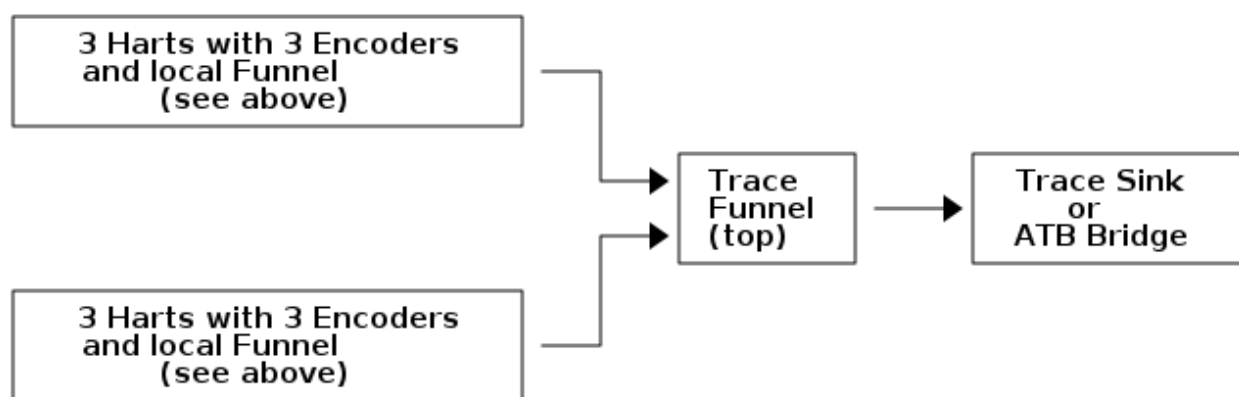


Figure 3. Multi-cluster trace: two three-hart clusters with top-level Funnel and Sink/Bridge

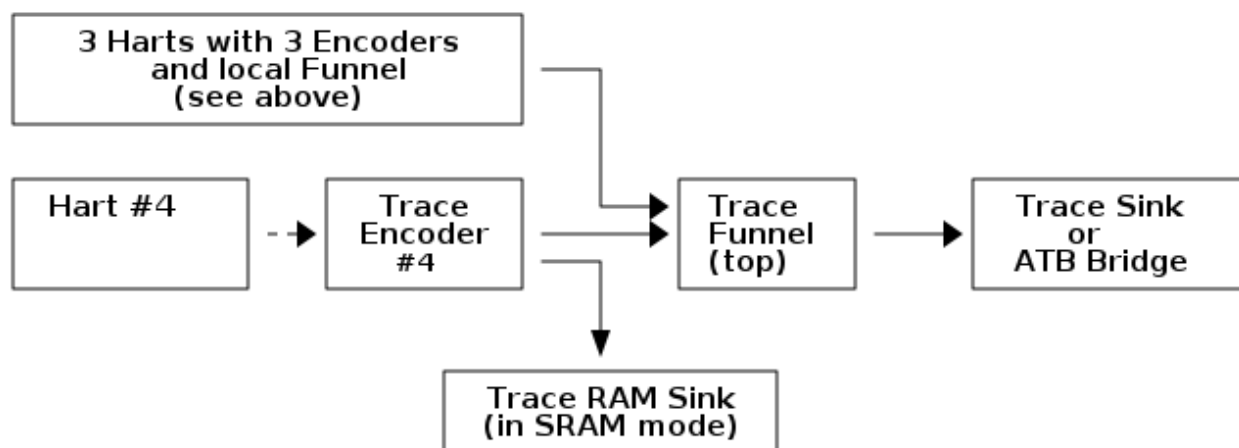


Figure 4. Local RAM Sink: Three-hart cluster plus extra hart with own RAM Sink (in SRAM mode)



Trace data from **Trace Encoder #4** may be combined with trace from other 3 Trace Encoders. But it may be also sent to dedicated **Trace RAM Sink** - in such a case

corresponding input to Trace Funnel (top) should be disabled.

4.2. Accessing Trace Control Registers

For the access method to the trace control registers, it makes a difference whether these registers shall be accessed by an external debug/trace tool, or by an internal debugger running on the chip.

Trace control register access by an external debugger (this is the most common use case):

- External debuggers must be able to access all trace control registers independent of whether the traced harts are running or halted. That is why for external debuggers, the recommended access method for memory-mapped control registers is memory accesses through the RISC-V debug module using SBA (System Bus Access) as defined in the RISC-V Debug Specification.

Trace control register access by an internal debugger:

- Through loads and stores performed by one or more harts in the system. Mapping the control interface into physical memory accessible from a hart allows that hart to manage a trace session independently from an external debugger. A hart may act as an internal debugger or may act in cooperation with an external debugger. Two possible use models are collecting crash information in the field and modifying trace collection parameters during execution. If a system has physical memory protection (PMP), a range can be configured to restrict access to the trace system from hart(s).



Additional control path(s) may also be implemented, such as extra JTAG registers or devices, a dedicated DMI debug bus or message-passing network. Such an access (which is NOT based on System Bus) may require custom implementation by trace probe vendors as this specification only mandates probe vendors to provide access via SBA commands.

4.3. Trace Component Register Map

Each block of 32-bit registers (for each component) has the following layout:

Table 3. Register Layout for Component

Address Offset	Register Name	Compliance	Description
0x000	tr??Control	Required	Main control register for this trace component
0x004	tr??Impl	Required	Trace Implementation information for this trace component
0x008 - 0x00F	extra controls	Optional	Extra controls for this trace component (named differently)
0x010 - 0xDFF	—	Optional	Additional registers (specific for the type of a component). All not used registers are reserved and should read as 0 and ignore writes.
0xE00 - 0xFFF	—	Optional	Registers reserved for implementation/vendor specific details. May allow identification of components on a system bus.



Each component has a tr??Active bit in the tr??Control register. Accesses to other registers are unspecified when the tr??Active bit is 0.

Each trace component has a tr??Impl register (at address offset 0x4) allowing trace component version and trace component type to be identified. This register allows debug tools to confirm the component type and potentially adjust tool behavior by looking at component versions.



Each component may have a different version. Initial version of this specification defines

all components to specify component version as 1.0 (major=1, minor=0).

Registers in the 4KB range that are not implemented are reserved and read as 0 and ignore writes.

Most trace control registers are optional. Some WARL fields may be hard coded to any value (including 0). It allows different implementations to provide different functionality.

Both N-Trace and E-Trace encoders are controlled by the same set of bits/fields in the same **trTe???** registers - as almost every register, field, bit is optional this provides good flexibility in implementation.

All other trace components are shared between different trace encoders (N-Trace and E-Trace).

4.3.1. Summary of Trace Encoder Registers

Table 4. Trace Encoder Registers (**trTe??**, **trTs??**)

Address Offset	Register Name	Compliance	Description
0x000	trTeControl	Required	Trace Encoder control register
0x004	trTeImpl	Required	Trace Encoder implementation information
0x008	trTeInstFeatures	Optional	Extra instruction trace encoder features and trace source IDs
0x00C	trTeInstFilters	Optional	Mask of filters to qualify an instruction trace
Data trace control (trTeData??)			
0x010	trTeDataControl	Optional	Data trace control and features
0x014 - 0x018	—	Reserved	Reserved for data trace related future standard extension
0x01C	trTeDataFilters	Optional	Mask of filters to qualify data trace
Reserved			
0x020 - 0x03F	—	Reserved	Reserved for future standard extension
Timestamp control (trTs??)			
0x040	trTsControl	Optional	Timestamp control register
0x044	—	Reserved	Reserved for future timestamp related standard extension
0x048	trTsCounterLow	Optional	Lower 32 bits of timestamp counter
0x04C	trTsCounterHigh	Optional	Upper bits of timestamp counter
Trigger control (trTeTrig??)			
0x050	trTeTrigDbgControl	Optional	Debug Triggers control register
0x054	trTeTrigExtInControl	Optional	External Triggers Input control register
0x058	trTeTrigExtOutControl	Optional	External Triggers Output control register
Reserved			
0x060 - 0x3FF	—	Reserved	Reserved for future standard extension
Filters & comparators (trTeFilter??, trTeComp??)			
0x400 - 0x5FF	trTeFilter??	Optional	Trace Encoder Filter Registers
0x600 - 0x7FF	trTeComp??	Optional	Trace Encoder Comparator Registers

4.3.2. Summary of Trace RAM Sink Registers

Table 5. Trace RAM Sink Registers (trRam??)

Address Offset	Register Name	Compliance	Description
0x000	trRamControl	Required	RAM Sink control register
0x004	trRamImpl	Required	RAM Sink Implementation information
0x008 - 0x00F	—	Reserved	Reserved for more control registers
0x010	trRamStartLow	Required	Lower 32 bits of start address of circular trace buffer
0x014	trRamStartHigh	Optional	Upper bits of start address of circular trace buffer
0x018	trRamLimitLow	Required	Lower 32 bits of end address of circular trace buffer
0x01C	trRamLimitHigh	Optional	Upper bits of end address of circular trace buffer
0x020	trRamWPLow	Required	Lower 32 bits of current write location for trace data in circular buffer
0x024	trRamWPHigh	Optional	Upper bits of current write location for trace data in circular buffer
0x028	trRamRPLow	Optional	Lower 32 bits of access pointer for trace readback
0x02C	trRamRPHigh	Optional	Upper bits of access pointer for trace readback
0x030 - 0x03F	—	Reserved	Reserved for more control registers
0x040	trRamData	Optional	Read/write access to SRAM trace memory (32-bit data)

4.3.3. Summary of Trace PIB Sink Registers

Table 6. Trace PIB Sink Registers (trPib??)

Address Offset	Register Name	Compliance	Description
0x000	trPibControl	Required	Trace PIB Sink control register
0x004	trPibImpl	Required	Trace PIB Sink Implementation information

4.3.4. Summary of Trace Funnel Registers

Table 7. Trace Funnel Registers (trFunnel??, trTs??)

Address Offset	Register Name	Compliance	Description
0x000	trFunnelControl	Required	Trace Funnel control register
0x004	trFunnelImpl	Required	Trace Funnel Implementation information
0x008	trFunnelDisInput	Optional	Disable individual funnel inputs
0x00C - 0x03F	—	Reserved	Reserved for more control registers
Timestamp control (trTs??)			
0x040	trTsControl	Optional	Timestamp control register
0x044	—	Reserved	Reserved for extra timestamp control
0x048	trTsCounterLow	Optional	Lower 32 bits of timestamp counter
0x04C	trTsCounterHigh	Optional	Upper bits of timestamp counter



Funnels may optionally be a source of timestamp and/or forward timestamp to Trace Encoders in the system. This way several Trace Encoders may share timestamp and trace from several harts may be time-correlated.

4.3.5. Summary of Trace ATB Bridge Registers

Table 8. Trace ATB Bridge Registers (trAtbBridge??)

Address Offset	Register Name	Compliance	Description
0x000	trAtbBridgeControl	Required	Trace ATB Bridge control register
0x004	trAtbBridgeImpl	Required	Trace ATB Bridge Implementation information

Chapter 5. Versioning of Components

Each component has a `tr??Impl` register, which includes two 4-bit `tr??VerMinor` and `tr??VerMajor` fields. These fields are guaranteed to be present in all future revisions of a standard, so trace tools will be able to discover a component version and act accordingly.

- Value 0 as `tr??VerMajor` is NOT allowed (due to compatibility reasons).
- Different components may report different versions (as some components may be updated more often than others).
- The major version `tr??VerMajor` field is incremented when the modification breaks backward compatibility.
- The minor version `tr??VerMinor` field is incremented when the modification maintains backward compatibility (for example adding a new field) - for that reason software should always write 0 to reserved bits in registers.
- Version 15.x is reserved for non-compatible version encoding.
- Version n.15 should be used as experimental (in development) implementation.

Software tools must report the version number as two decimal numbers *major.minor* - initial version of this specification is defined as **1.0**.



Trace software should handle versions as follows (let's assume hypothetical version 2.3 was defined as current version in moment of release of trace software)

- *0.x ⇒ Reject as not supported or generate a warning and handle as pre-ratified/initial version 0.*
- *2.3 ⇒ Accept silently.*
- *2.2 ⇒ Accept silently (and trim features or not allow users to set newer features).*
- *2.4 ⇒ Generate a warning but continue using 2.3 features.*
- *2.15 ⇒ Generate an "experimental version" warning but continue using 2.3 features.*
- *1.x ⇒ Generate a warning and continue or reject as an obsolete (referring to last debugger supporting this version).*
- *3.x ⇒ Generate a fatal error that this future version is not compatible with existing software and possibly redirect to the tool update page.*

Displayed messages should report component name, component base address and current and supported version numbers. It is suggested to display the full hexadecimal value of `tr??Impl` register as it may aid in debugging of possibly incorrect/incompatible component configuration.

Chapter 6. Trace Encoder Control Interface

Many features of the Trace Encoder (TE for short) are optional. In most cases, optional features are enabled using a WARL (write any, read legal) register field. A debugger can determine if optional feature is present by writing to the register field and reading back the result.

Table 9. Register: trTeControl: Trace Encoder Control Register (trBaseEncoder+0x000)

Bit	Field	Description	RW	Reset
0	trTeActive	Primary activate/reset bit for the TE. When 0, the TE may have clocks gated off or be powered down, and other register locations may be inaccessible. Hardware may take an arbitrarily long time to process power-up and power-down and will indicate completion when the read value of this bit matches what was written. See Reset and Discovery chapter for more details.	RW	0
1	trTeEnable	1: Trace Encoder is enabled. Allows trTeInstTracing and trTeDataTracing to turn tracing on and off. Setting trTeEnable to 0 flushes any queued trace data to the sink or funnel attached to this encoder. This bit can be set to 1 only by direct writing to it. This write of 1 should be done after all other settings are done. See Enabling and Disabling chapter for more details.	RW	0
2	trTeInstTracing	1: Instruction trace is being generated. Written from a trace tool (after a write to trTeEnable) or controlled by triggers. When trTeInstTracing=1 , instruction trace data may be subject to additional filtering in some implementations (additional trTeInstMode settings).	RW	Undef
3	trTeEmpty	Reads as 1 when all generated trace have been emitted.	RO	1
6:4	trTeInstMode	Instruction trace generation mode 0: Full Instruction trace is disabled, but other trace (data trace) may be emitted. 1-2: Protocol defined trace mode. 3: Baseline instruction trace (for example Branch Trace). 4-5: Protocol defined trace mode. 6: Optimized instruction trace (for example Branch History Trace). 7: Reserved for vendor-defined instruction trace mode. NOTE: When non-supported mode (different than 0) is set, it cannot revert to 0 but MUST revert to supported non-0 mode.	WARL	Undef
8:7	—	Reserved	—	0
9	trTeContext	Enable sending trace messages/fields with scontext/mcontext values and/or privilege levels.	WARL	Undef
10	—	Reserved	—	0
11	trTeInstTrigEnable	1: Allows trTeInstTracing to be set or cleared by Trace-on and Trace-off signals generated by the corresponding trigger module.	WARL	Undef
12	trTeInstStallOrOverflow	Set to 1 by hardware when trace buffer overflow (also known as trace lost) occurs, or when the TE requests a hart stall. Clears to 0 at TE reset or when the trace is enabled (trTeEnable set to 1). Write 1 to clear.	RW1C	Undef
13	trTeInstStallEna	0: If TE cannot send a message, the message is dropped. The protocol dependent overflow instruction trace synchronization message/packet is generated when the trace is restarted, so the decoder will know that trace is lost and must reset any internal decoder state. 1: If TE cannot send a message, the hart is stalled until it can. With this option execution of instructions by the hart may be intrusively affected, but in many cases it is acceptable.	WARL	Undef
14	—	Reserved	—	0
15	trTeInhibitSrc	0: Messages/packets generated by the trace encoder include a message source field if the source width held in trTeSrcBits is not 0. 1: Disable inclusion of source field in trace messages/packets.	WARL	Undef

Bit	Field	Description	RW	Reset
17:16	trTeInstSyncMode	Select the periodic instruction trace synchronization message/packet generation mechanism. At least one non-zero mechanism must be implemented. 0: Off 1: Count trace messages/packets 2: Count hart clock cycles 3: Count instruction 16-bit half-words Once the max value of periodic counter is reached, an instruction trace synchronization message/packet should be generated.	WARL	Undef
19:18	—	Reserved	—	0
23:20	trTeInstSyncMax	The maximum interval (in units determined by trTeInstSyncMode) between instruction trace synchronization messages/packets. Generate synchronization when count reaches $2^{(\text{trTeInstSyncMax}+4)}$. If an instruction trace synchronization message/packet is generated for another reason, the internal counter should be reset.	WARL	Undef
26:24	trTeFormat	Trace recording/protocol format: 0: Format defined by Efficient Trace for RISC-V (E-Trace) Specification 1: Format defined by RISC-V N-Trace (Nexus-based Trace) Specification 2-6: Reserved for future formats 7: Vendor-specific format	WARL	Undef
31:27	—	Reserved	—	0



Writing to this register while trace is enabled may unintentionally change a value of **trTeInstTracing** bit because that bit may dynamically change by triggers.

Table 10. Register: **trTeImpl**: Trace Encoder Implementation Register (**trBaseEncoder+0x004**)

Bit	Field	Description	RW	Reset
3:0	trTeVerMajor	Trace Encoder Component Major Version. Value 1 means the component is compliant with this document. Value 0 means pre-ratified/initial version - see 'Pre-ratified/Initial Interface Version' chapter at the end.	RO	1
7:4	trTeVerMinor	Trace Encoder Component Minor Version. Value 0 means the component is compliant with this document.	RO	0
11:8	trTeCompType	Trace Encoder Component Type (Trace Encoder)	RO	0x1
15:12	—	Reserved for future versions of this standard	—	0
19:16	trTeProtocolMajor	Trace Protocol Major Version. As specified by specification governing trTeFormat .	RO	SD
23:20	trTeProtocolMinor	Trace Protocol Minor Version. As specified by specification governing trTeFormat .	RO	SD
31:24	—	Reserved for vendor specific implementation details	—	SD



trTeProtocol?? fields are separated from **trTeVer??** as we may have the same control interface, but protocol itself may be extended with new packets/ messages/ fields.

Table 11. Register: **trTeInstFeatures**: Trace Instruction Features Register (**trBaseEncoder+0x008**)

Bit	Field	Description	RW	Reset
0	trTeInstNoAddrDiff	When set, trace messages/packets always carry a full address.	WARL	Undef
1	trTeInstNoTrapAddr	When set, do not include trap handler address in trap messages/packets.	WARL	Undef

Bit	Field	Description	RW	Reset
2	trTeInstEnSequentialJump	When set, treat sequentially inferrable jumps as inferable PC discontinuities.	WARL	Undef
3	trTeInstEnImplicitReturn	When set, treat returns as inferable PC discontinuities when returning from a recent call on a stack. Field trTeInstImplicitReturnMode below provides more details.	WARL	Undef
4	trTeInstEnBranchPrediction	When set, Branch Predictor based compression is enabled.	WARL	Undef
5	trTeInstEnJumpTargetCache	When set, Jump Target Cache based compression is enabled.	WARL	Undef
7:6	trTeInstImplicitReturnMode	Defines how the decoder is handling stack of return addresses (if enabled by trTeInstEnImplicitReturn bit): 0 : Implicit Return mode is not supported, or implementation is not reporting how it is implemented. 1 : Simple level counting without the return address comparing. 2 : Partial (LSB portion of return address) compare (smaller logic cost than 3 below, but in most cases adequate as chances to have an incorrect return address with same LSB bits is very slim). 3 : Full address comparing (always assures skipped return addresses are the same as addresses deducted from call instruction). Implementation may take advantage of RAS (Return Address Stack) if implemented by the hart.	WARL	Undef
8	trTeInstEnRepeatedHistory	Enable repeated branch history/map detection when set.	WARL	Undef
9	trTeInstEnAllJumps	Enable emitting of trace message or add history/map bit for direct unconditional/inferable control flow changes (jumps or calls). Normally these instructions do not generate any trace as the decoder can determine the next instruction. Trace will not compress well but timestamp accuracy will be better - may be used when profiling loops.	WARL	Undef
10	trTeInstExtendAddrMSB	When set, allow extended handing of MSB address bits. Encoding details are trace protocol dependent.	WARL	Undef
15:11	—	Reserved for additional instruction trace control/status bits	—	0
27:16	trTeSrcID	Trace source ID assigned to this trace encoder. If trTeSrcBits is not 0 and trace source is not disabled by trTeInhibitSrc , then trace messages from this TE will all include a trace source field of trTeSrcBits bits and all messages from this TE will use this value as trace source field.	WARL	Undef
31:28	trTeSrcBits	The number of bits in the trace source field (0..12), unless disabled by trTeInhibitSrc . Some trace protocols may require that this field is identical for all enabled trace encoders within the same trace stream.	WARL	Undef



Applicability of different **trTeInst??** fields for each trace encoding protocol is described in a document which defines the protocol (and not all fields are applicable to all protocols).

Table 12. Register: **trTeInstFilters**: Trace Instruction Filters Register (**trBaseEncoder+0x00C**)

Bit	Field	Description	RW	Reset
15:0	trTeInstFilters	Determine which filters defined in Trace Encoder Filter Registers chapter qualify an instruction trace. If bit <i>n</i> is a 1 then instructions will be traced when filter <i>n</i> matches. If all bits are 0, all instructions are traced.	WARL	Undef
31:16	—	Reserved	—	0

Table 13. Register: trTeDataControl: Data Trace Control Register (trBaseEncoder+0x010)

Bit	Field	Description	RW	Reset
0	trTeDataImplemented	Read as 1 if data trace is implemented.	RO	SD
1	trTeDataTracing	1: Data trace is being generated. Written from a trace tool or controlled by triggers. When trTeDataTracing =1, data trace may be subject to additional filtering in some implementations.	WARL	Undef
2	trTeDataTrigEnable	Global enable/disable for data trace triggers	WARL	Undef
3	trTeDataStallOrOverflow	Set to 1 by hardware when data trace causes trace buffer overflow, or when the TE requests a hart stall due to data trace. Clears to 0 at TE reset or when the trace is enabled (trTeEnable set to 1). Write 1 to clear.	RW1C	Undef
4	trTeDataStallEna	0: If TE cannot send data trace messages, an overflow message is generated when the trace is restarted. 1: If TE cannot send data trace messages, the hart is stalled until it can.	WARL	Undef
5	trTeDataDrop	Written to 1 by hardware when the data trace packet was dropped (if enabled). Clears to 0 at TE reset or when the trace is enabled (trTeEnable set to 1). Write 1 to clear.	RW1C	Undef
6	trTeDataDropEna	1: Allow temporary suppression of data trace (at some watermark level) to prevent trace overflow or stall. This way instruction trace will have higher priority.	WARL	Undef
15:7	—	Reserved for additional data trace control/status bits.	—	0
16	trTeDataNoValue	When set, omit data values from data trace packets.	WARL	Undef
17	trTeDataNoAddr	When set, omit data address from data trace packets.	WARL	Undef
19:18	trTeDataAddrCompress	Data trace address compression selection: 0: Only send full (unmodified) addresses 1: Use XOR compression 2: Use differential compression 3: Protocol defined address compression	WARL	Undef
31:20	—	Reserved	—	0



Writing to this register while trace is enabled may unintentionally change a value of **trTeDataTracing** bit because that bit may dynamically change by triggers.



Applicability of different **trTeData??** fields for each trace encoding protocol is described in a document which defines the protocol (and not all fields are applicable to all protocols).

Table 14. Register: trTeDataFilters: Trace Data Filters Register (trBaseEncoder+0x01C)

Bit	Field	Description	RW	Reset
15:0	trTeDataFilters	Determine which filters defined in Trace Encoder Filter Registers chapter qualify data trace. If bit <i>n</i> is a 1 then data accessed will be traced when filter <i>n</i> matches. If all bits are 0, all data accesses are traced.	WARL	Undef
31:16	—	Reserved	—	0

6.1. Timestamp Unit

Timestamp Unit is an optional sub-component present in either Trace Encoder or Trace Funnel. An implementation may choose from several modes of timestamps:

- **Internal System** - fixed clock in a system (such as bus clock) is used to increment the timestamp counter (for both Trace Encoders and Trace Funnels)
- **Internal Core** - core clock is used to increment the timestamp counter (only for Trace Encoders)
- **Shared** - shares timestamp with another Trace Encoder or Trace Funnel
- **External** - accepts a binary timestamp value from an outside source such as ARM CoreSight™ trace (for both Trace Encoders and Trace Funnels)

Implementations may have no timestamp, one timestamp mode, or more than one mode. The WARL field **trTsMode** is used to determine the system capability and to set the desired timestamp mode.

The width of the timestamp is implementation dependent, typically 40 or 48 bits (40-bit timestamp will overflow every 4.7 minutes assuming 1GHz timestamp clock).

In a system with Funnels, typically all the Funnels are built with a Timestamp Unit. The top-level Funnel is the source of the timestamp (Internal System or External) and all the Encoders and other Funnels have a Shared timestamp. This assures that all timestamps in the system are the same and trace from different harts may be time-correlated. To perform the forwarding function, the mid-level Funnels must be programmed with **trFunnelActive** = 1 (which is natural as all trace messages must pass through that funnel).

An Internal System or Core timestamp unit may include a timestamp clock pre-scaler divider, which can extend the range of a narrower timestamp and uses less power but has less resolution.

In a system with an Internal Core timestamp counter (implemented in Trace Encoder associated with a hart) an optional control bit is provided to stop the counter when the hart is halted by a debugger.

Table 15. Register: trBaseEncoder/Funnel+0x040 trTsControl: Timestamp Control Register

Bit	Field	Description	RW	Reset
0	trTsActive	Primary activate/reset bit for timestamp unit. This must either be RW or, if separated reset for timestamp component is not implemented, a read-only copy of the corresponding trTeActive or trFunnelActive bit. See Reset and Discovery chapter for more details.	WARL	SD
1	trTsCount	Internal System or Core timestamp only. 1: counter runs, 0: counter stopped.	WARL	Undef
2	trTsReset	Internal System or Core timestamp only. Write 1 to reset the timestamp counter.	W1	—
3	trTsRunInDebug	Internal Core timestamp only. 1: counter runs when hart is halted (in debug mode), 0: stopped	WARL	Undef
6:4	trTsMode	Mode used by Timestamp unit: 0: None 1: External 2: Internal System 3: Internal Core 4: Shared 5-7: Vendor-specific mode	WARL	Undef
7	—	Reserved	—	0

Bit	Field	Description	RW	Reset
9:8	trTsPrescale	Internal System or Core timestamp only. Prescale timestamp input clock by $2^{(2 \cdot \text{trTsPrescale})}$. It will be divided by 1, 4, 16, 64 respectively.	WARL	Undef
14:10	—	Reserved	—	0
15	trTsEnable	Enable for timestamp field in trace messages/packets (for Trace Encoder only).	WARL	Undef
23:16		Vendor-specific bits to control what message/packet types include timestamp fields.	WARL	Undef
29:24	trTsWidth	Width of timestamp in bits (0..63)	RO	SD
31:30	—	Reserved	—	0

Table 16. Register: trTsCounterLow: Timestamp Counter Lower Bits (trBaseEncoder/Funnel+0x048)

Bit	Field	Description	RW	Reset
31:0	trTsCounterLow	Lower 32 bits of timestamp counter.	RO	0

Table 17. Register: trTsCounterHigh: Timestamp Counter Upper Bits (trBaseEncoder/Funnel+0x04C)

Bit	Field	Description	RW	Reset
31:0	trTsCounterHigh	Upper bits of timestamp counter, zero-extended.	RO	0

6.2. Trace Encoder Triggers

6.2.1. Debug Trigger Module

Debug module triggers are signals from the hart that a trigger was hit, but the action associated with that trigger is a trace-related action. Action identifiers 2-5 are reserved for trace actions in the RISC-V Debug Specification, where triggers are defined. Actions 2-4 are defined by the Efficient Trace for RISC-V (E-Trace) Specification. The desired action is written to the **action** field of the Match Control **mcontrol** CSR (0x7a1). As not all harts may support all trace actions, the debugger should read back the **mcontrol** CSR after setting the desired trace action to verify that the option exists.

Table 18. Debug Trigger Actions

Trigger Action (from debug spec)	Effect
0	Breakpoint exception (as defined in RISC-V Debug Specification)
1	Debug exception (as defined in RISC-V Debug Specification)
2	Trace-on action When trTeInstTrigEnable = 1 it will start instruction tracing (trTeInstTracing → 1). When trTeDataTrigEnable = 1 it will start data tracing (trTeDataTracing → 1).
3	Trace-off action When trTeInstTrigEnable = 1 it will stop instruction tracing (trTeInstTracing → 0). When trTeDataTrigEnable = 1 it will stop data tracing (trTeDataTracing → 0).
4	Trace-notify action If tracing is active (trTeInstTracing = 1), then the encoder generates a packet with the current PC and, if enabled, a timestamp.
5	Vendor-specific trace action (optional)

If there are vendor-specific features that require control, the **trTeTrigDbgControl** register is used.

Table 19. Register: **trTeTrigDbgControl**: Debug Trigger Control Register (**trBaseEncoder**+0x050)

Bit	Field	Description	RW	Reset
31:0	trTeTrigDbgControl	Vendor-specific trigger setup	WARL	Undef

6.2.2. External Trace Triggers

The TE may be configured with up to 8 external trigger inputs for controlling trace. These are in addition to the external triggers present in the Debug Module when Halt Groups are implemented. The specific hardware signals comprising an external trigger are implementation dependent.

External Trigger Outputs may also be present. A trigger out may be generated by trace starting, trace stopping, a watchpoint, or by other system-specific events.

Table 20. Register: trTeTrigExtInControl: External Trigger Input Control Register (trBaseEncoder+0x054)

Bit	Field	Description	RW	Reset
3:0	trTeTrigExtInAction0	Select action to perform when external trigger input #0 fires. If external trigger input #0 does not exist, then its action is fixed at 0. 0: No action 1: Reserved 2: Trace-on action When <code>trTeInstTrigEnable</code> = 1 it will start instruction tracing (<code>trTeInstTracing</code> → 1). When <code>trTeDataTrigEnable</code> = 1 it will start data tracing (<code>trTeDataTracing</code> → 1). 3: Trace-off action When <code>trTeInstTrigEnable</code> = 1 it will stop instruction tracing (<code>trTeInstTracing</code> → 0). When <code>trTeDataTrigEnable</code> = 1 it will stop data tracing (<code>trTeDataTracing</code> → 0). 4: Trace-notify action If tracing is active (<code>trTeInstTracing</code> = 1), then the encoder generates a packet with the current PC and, if enabled, a timestamp. 5-15: Reserved	WARL	Undef
31:4	trTeTrigExtInActionN	Select actions (as defined for bits 3-0) for external trigger input #N (1..7). If an external trigger input does not exist, then its action is fixed at 0.	WARL	Undef

Table 21. Register: trTeTrigExtOutControl: External Trigger Output Control Register (trBaseEncoder+0x058)

Bit	Field	Description	RW	Reset
3:0	trTeTrigExtOutEvent0	Bitmap to select which event(s) cause external trigger #0 output to fire. If external trigger output #0 does not exist, then all bits are fixed at 0. Bits 2 and 3 may be fixed at 0 if the corresponding feature is not implemented. Bit 0: Start trace transition (<code>trTeInstTracing</code> 0 → 1) will fire the trigger. Bit 1: Stop trace transition (<code>trTeInstTracing</code> 1 → 0) will fire the trigger. Bit 2-3: Vendor-specific event (optional)	WARL	Undef
31:4	trTeTrigExtOutEventN	Select events for external trigger output #N (1..7). If an external trigger output does not exist, then its event bits are fixed at 0	WARL	Undef

6.2.3. Triggers Precedence

It is implementation dependent what happens when triggers (from debug module or external) with conflicting actions occur simultaneously (signaled at the same ingress port cycle) or if triggers occur too frequently. It is recommended that tracing starts from the oldest instruction retired in the cycle that Trace-on is asserted, and stops following the newest instruction retired in the cycle that Trace-off is asserted.

6.3. Trace Encoder Filter Registers

All registers with offsets 0x400 .. 0x7FC are designated for additional trace encoder filter options (context, addresses, modes, etc.).

Trace encoder filters are an optional feature that can be used to control the generated trace in various ways.

The registers below divide the filter logic into filters and comparators to provide maximum flexibility at low cost. The number of filters and comparators depends on the system. Each filter unit can specify filtering against instruction and optionally against data trace inputs from the hart. When filter i is implemented, the registers `trTeFilteriControl` and `trTeInstFilters` must be implemented to enable it. And to apply filter i to the data trace, the `trTeDataFilters` register must also be present. And if a match bit in the `trTeFilteriControl` register can be set to 1 (= enabling a filter option), the corresponding register from the bit's description must have a correct value already set as otherwise the trigger may fire unintentionally. Each of the mentioned comparator units is a pair of comparators (primary and secondary, or P and S), so a limited range can be matched with a single comparator unit if needed. Each enabled filter define independent condition where trace is enabled - if several filters are enabled they act as **logical OR**. Several conditions for single filter act as **logical AND**.



Filter and comparator registers refer to values of some signals (as `priv`, `itype`, `ecause`, `dtype`, `dsize`, ...) available on Trace Ingress Port. See E-Trace specification for details of encoding of these values.

Table 22. Register: `trTeFilter??`: Trace Encoder Filter Registers (`trBaseEncoder+0x400..0x5FF`)

Address Offset	Register Name	Compliance	Description
$0x400 + 0x20*i$	<code>trTeFilteriControl</code>	Optional	Filter i control
$0x404 + 0x20*i$	<code>trTeFilteriMatchInst</code>	Optional	Filter i instruction match control
$0x408 + 0x20*i$	<code>trTeFilteriMatchEcauseLow</code>	Optional	Filter i Ecause match control (bits 31:0)
$0x40C + 0x20*i$	<code>trTeFilteriMatchEcauseHigh</code>	Optional	Filter i Ecause match control (bits 63:32)
$0x410 + 0x20*i$	<code>trTeFilteriMatchValueImpdef</code>	Optional	Filter i impdef value
$0x414 + 0x20*i$	<code>trTeFilteriMatchMaskImpdef</code>	Optional	Filter i impdef mask
$0x418 + 0x20*i$	<code>trTeFilteriMatchData</code>	Optional	Filter i Data trace match control
$0x41C + 0x20*i$	—	Optional	Reserved

Table 23. Register: `trTeComp??`: Trace Encoder Comparator Registers (`trBaseEncoder+0x600..0x6FF`)

Address Offset	Register Name	Compliance	Description
$0x600 + 0x20*j$	<code>trTeCompjControl</code>	Optional	Comparator j control
$0x604 + 0x20*j$	—	Optional	Reserved
$0x608 + 0x20*j$	—	Optional	Reserved
$0x60c + 0x20*j$	—	Optional	Reserved
$0x610 + 0x20*j$	<code>trTeCompjPmatchLow</code>	Optional	Comparator j primary match (bits 31:0)
$0x614 + 0x20*j$	<code>trTeCompjPmatchHigh</code>	Optional	Comparator j primary match (bits 63:32)
$0x618 + 0x20*j$	<code>trTeCompjSmatchLow</code>	Optional	Comparator j secondary match (bits 31:0)
$0x61C + 0x20*j$	<code>trTeCompjSmatchHigh</code>	Optional	Comparator j secondary match (bits 63:32)

Table 24. Register: **trTeFilteriControl** : Filter *i* Control Register (**trBaseEncoder+0x400 + 0x20i**)

Bit	Field	Description	RW	Reset
0	trTeFilterEnable	Overall filter enable for filter # <i>i</i>	WARL	Undef
1	trTeFilterMatchPrivilege	When set, match privilege levels specified by trTeFilterMatchChoicePrivilege field for filter # <i>i</i> .	WARL	Undef
2	trTeFilterMatchEcause	When set, start matching from exception cause codes specified by trTeFilterMatchChoiceEcause field for filter # <i>i</i> , and stop matching upon return from the 1st matching exception.	WARL	Undef
3	trTeFilterMatchInterrupt	When set, start matching from either an interrupt or exception as specified by trTeFilterMatchValueInterrupt field for filter # <i>i</i> , and stop matching upon return from the 1st matching trap.	WARL	Undef
4	trTeFilterMatchComp1	When set, the output of the comparator selected by trTeFilterComp1 must be true for the filter to match.	WARL	Undef
7:5	trTeFilterComp1	Specifies the comparator unit to use for the 1st comparison.	WARL	Undef
8	trTeFilterMatchComp2	When set, the output of the comparator selected by trTeFilterComp2 must be true for the filter to match.	WARL	Undef
11:9	trTeFilterComp2	Specifies the comparator unit to use for the 2nd comparison.	WARL	Undef
12	trTeFilterMatchComp3	When set, the output of the comparator selected by trTeFilterComp3 must be true for the filter to match.	WARL	Undef
15:13	trTeFilterComp3	Specifies the comparator unit to use for the 3rd comparison.	WARL	Undef
16	trTeFilterMatchImpdef	When set, match impdef values as specified by trTeFilterMatchValueImpdef and trTeFilterMatchMaskImpdef fields for filter # <i>i</i> .	WARL	Undef
23:17	—	Reserved	—	0
24	trTeFilterMatchDtype	When set, match dtype values as specified by trTeFilterMatchChoiceDtype field for filter # <i>i</i> .	WARL	Undef
25	trTeFilterMatchDsize	When set, match dsize values as specified by trTeFilterMatchChoiceDsize field for filter # <i>i</i> .	WARL	Undef
31:26	—	Reserved	—	0



Handling of **trTeFilterMatchEcause** and **trTeFilterMatchInterrupt** should include a count of nested traps. The size of the counter is implementation dependent. If the number of nested traps exceeds the number that can be counted, the counter will saturate, meaning that the filtering will turn off prematurely.

Table 25. Register: **trTeFilteriMatchInst** : Filter *i* Instruction Match Control Register (**trBaseEncoder+0x404 + 0x20i**)

Bit	Field	Description	RW	Reset
7:0	trTeFilterMatchChoicePrivilege	When trTeFilterMatchPrivilege field for filter # <i>i</i> is set, match all privilege levels for which the corresponding bit is set. For example, if bit <i>N</i> is 1, then match if the priv value at ingress port is <i>N</i> . Setting several bits allow matching several privileges.	WARL	Undef
8	trTeFilterMatchValueInterrupt	When trTeFilterMatchInterrupt field for filter # <i>i</i> is set, match itype of 2 or 1 depending on whether this bit is 1 or 0 respectively.	WARL	Undef
31:9	—	Reserved	—	0

Table 26. Register: trTeFilteriMatchEcauseLow : Filter *i* Ecause Match Control (low) Register (trBaseEncoder+0x408 + 0x20i)

Bit	Field	Description	RW	Reset
31:0	trTeFilterMatchChoiceEcauseLow	When trTeFilterMatchEcause field for filter # <i>i</i> is set, match all exception causes for which the corresponding bit is set. If bit <i>N</i> is 1, then match if the ecause is <i>N</i> .	WARL	Undef

Table 27. Register: trTeFilteriMatchEcauseHigh : Filter *i* Ecause Match Control (high) Register (trBaseEncoder+0x40C + 0x20i)

Bit	Field	Description	RW	Reset
31:0	trTeFilterMatchChoiceEcauseHigh	Stores bits 63:32 to allow matching of higher ecause codes. If bit <i>N</i> is 1, then match if the ecause is <i>N</i> +32.	WARL	Undef

Table 28. Register: trTeFilteriMatchValueImpdef : Filter *i* Impdef Match Value Register (trBaseEncoder+0x410 + 0x20i)

Bit	Field	Description	RW	Reset
31:0	trTeFilterMatchValueImpdef	When trTeFilterMatchimpdef field for filter # <i>i</i> is set, match if (impdef & trTeFilterMatchMaskImpdef) == (trTeFilterMatchValueImpdef & trTeFilterMatchMaskImpdef).	WARL	Undef

Table 29. Register: trTeFilteriMatchMaskImpdef : Filter *i* Impdef Match Mask Register (trBaseEncoder+0x414 + 0x20i)

Bit	Field	Description	RW	Reset
31:0	trTeFilterMatchMaskImpdef	When trTeFilterMatchimpdef field for filter # <i>i</i> is set, match if (impdef & trTeFilterMatchMaskImpdef) == (trTeFilterMatchValueImpdef & trTeFilterMatchMaskImpdef).	WARL	Undef

Table 30. Register: trTeFilteriMatchData : Filter *i* Data Match Control Register (trBaseEncoder+0x418 + 0x20i)

Bit	Field	Description	RW	Reset
15:0	trTeFilterMatchChoiceDtype	When trTeFilterMatchDtype field for filter # <i>i</i> is set, match all data access types for which the corresponding bit is set. For example, if bit <i>N</i> is 1, then match if the dtype value is <i>N</i> .	WARL	Undef
23:16	trTeFilterMatchChoiceDsize	When trTeFilterMatchDsize field for filter # <i>i</i> is set, match all data access sizes for which the corresponding bit is set. For example, if bit <i>N</i> is 1, then match if the dsize value is <i>N</i> .	WARL	Undef
31:24	—	Reserved	—	0

Table 31. Register: `trTeCompjControl` : Comparator j Control Register (`trBaseEncoder+0x600 + 0x20j`)

Bit	Field	Description	RW	Reset
1:0	<code>trTeCompPInput</code>	Determines which input to compare against the primary comparator. 0: <code>iaddr</code> 1: <code>context</code> 2: <code>tval</code> 3: <code>daddr</code>	WARL	Undef
3:2	<code>trTeCompSInput</code>	Determines which input to compare against the secondary comparator. Same encoding as <code>trTeCompPInput</code> .	WARL	Undef
6:4	<code>trTeCompPFunction</code>	Selects the primary comparator function. Primary result is true if input selected via <code>trTeCompPInput</code> is: 0: equal to <code>trTeCompPMatch</code> 1: not equal to <code>trTeCompPMatch</code> 2: less than <code>trTeCompPMatch</code> 3: less than or equal to <code>trTeCompPMatch</code> 4: greater than <code>trTeCompPMatch</code> 5: greater than or equal to <code>trTeCompPMatch</code> 6: Result always false (input ignored). Prime latch to 1 if <code>trTeCompMatchMode</code> is 3 7: Result always true (input ignored)	WARL	Undef
7	—	Reserved	—	0
10:8	<code>trTeCompSFunction</code>	Selects the secondary comparator function. Secondary result is true if input selected via <code>trTeCompSInput</code> is: 0: equal to <code>trTeCompSMatch</code> 1: not equal to <code>trTeCompSMatch</code> 2: less than <code>trTeCompSMatch</code> 3: less than or equal to <code>trTeCompSMatch</code> 4: greater than <code>trTeCompSMatch</code> 5: greater than or equal to <code>trTeCompSMatch</code> 6: Result always true (input ignored). Use <code>trTeCompSMatch</code> as a mask for <code>trTeCompPMatch</code> 7: Result always true (input ignored)	WARL	Undef
11	—	Reserved	—	0
13:12	<code>trTeCompMatchMode</code>	Selects the match condition used to assert the overall comparator output 0: primary result true 1: primary and secondary result both true: (P && S) 2: Either primary or secondary result does not match: !(P && S) 3: Set when primary result is true and continue to assert until instruction after secondary result is true	WARL	Undef
14	<code>trTeCompPNotify</code>	Generate a trace packet explicitly reporting the address of the final instruction in a block that causes a primary match. This is also known as a watchpoint. Requires <code>trTeCompPInput</code> to be 0, and has no effect otherwise.	WARL	Undef
15	<code>trTeCompSNotify</code>	Generate a trace packet explicitly reporting the address of the final instruction in a block that causes a secondary match. This is also known as a watchpoint. Requires <code>trTeCompSInput</code> to be 0, and has no effect otherwise.	WARL	Undef
31:16	—	Reserved	—	0



Comparisons are performed as unsigned numbers. Only bits from an input signal (as defined by `trTeCompPInput` and/or `trTeCompSInput` fields), should be compared. Additional most significant bits from the `trTeCompjPMatchLow/High` registers must be ignored.

**Table 32. Register: trTeCompjPMatchLow : Comparator j Primary match (low) Register
(trBaseEncoder+0x610 + 0x20j)**

Bit	Field	Description	RW	Reset
31:0	trTeCompPMatchLow	The match value for the primary comparator (bits 31:0).	WARL	Undef

**Table 33. Register: trTeCompjPMatchHigh : Comparator j Primary match (high) Register
(trBaseEncoder+0x614 + 0x20j)**

Bit	Field	Description	RW	Reset
31:0	trTeCompPMatchHigh	The match value for the primary comparator (bits 63:32).	WARL	Undef

**Table 34. Register: trTeCompjSMatchLow : Comparator j Secondary match (low) Register
(trBaseEncoder+0x618 + 0x20j)**

Bit	Field	Description	RW	Reset
31:0	trTeCompSMatchLow	The match value for the secondary comparator (bits 31:0).	WARL	Undef

**Table 35. Register: trTeCompjSMatchHigh : Comparator j Secondary match (high) Register
(trBaseEncoder+0x61C + 0x20j)**

Bit	Field	Description	RW	Reset
31:0	trTeCompSMatchHigh	The match value for the secondary comparator (bits 63:32).	WARL	Undef

Chapter 7. Trace RAM Sink

Trace RAM Sink may be instantiated or configured to support storing trace into dedicated SRAM or system memory. SRAM mode is using dedicated local memory inside of RAM sink, while system memory mode (SMEM mode) is accessing memory via system bus (care should be taken to not overwrite application code or data - it is usually done by reserving part of system memory for trace). Dedicated SRAM memory must be read via dedicated **trRamData** register, while memory in SMEM mode should be read as any other memory on system bus - for example using SBA (System Bus Access) access mode as defined in the RISC-V Debug Specification.

Trace data is placed in memory in LSB order (first byte of trace packet/data is placed on LSB).

Be aware that in case trace memory wraps around some protocols may require additional synchronization data - it is usually done by periodically generating a sequence of alignment synchronization bytes which cannot be part of any valid packet. Specification of each trace protocol must define it.

Table 36. Register: trRamControl: Trace RAM Sink Control Register (trBaseRam+0x000)

Bit	Field	Description	RW	Reset
0	trRamActive	Primary activate/reset bit for Trace RAM Sink. When 0, the Trace RAM Sink may have clocks gated off or be powered down, and other register locations may be inaccessible. Hardware may take an arbitrarily long time to process power-up and power-down and will indicate completion when the read value of this bit matches what was written. See Reset and Discovery chapter for more details.	RW	0
1	trRamEnable	1: Trace RAM Sink enabled. Setting trRamEnable to 0 flushes any queued trace data to memory (idle bytes/packet may be appended after the last message/packet to assure memory access alignment). See Enabling and Disabling chapter for more details. Enabling trace CANNOT change any of trRamStart/Limit/WP/RP?? registers. Disabling trace may update trRamWP?? because of flushing.	RW	0
2	—	Reserved	—	0
3	trRamEmpty	Reads 1 when Trace RAM Sink internal buffers are empty, which means that all trace data is flushed.	RO	1
4	trRamMode	0: This RAM Sink will operate in SRAM mode 1: This RAM Sink will operate in SMEM mode	WARL	Undef
7:5	—	Reserved	—	0
8	trRamStopOnWrap	1: Disable storing trace to RAM (trRamEnable → 0) when the circular buffer gets full. Sink should stop accepting new messages which may result in an overflow or stall condition at an encoder.	WARL	Undef
10:9	trRamMemFormat	0: Memory is formatted as plain bytes 1-2: Reserved for future formats 3: Reserved for custom memory format	WARL	Undef
11	—	Reserved	—	0
14:12	trRamAsyncFreq	0: Alignment synchronization (Async) packets disabled (may be the only choice for some protocols) 1-7: Different levels of alignment synchronization (bigger number, bigger distance). Details should be defined in the specification of each trace protocol.	WARL	Undef
31:15	—	Reserved	—	0

Table 37. Register: **trRamImpl**: Trace RAM Sink Implementation Register (**trBaseRamSink+0x004**)

Bit	Field	Description	RW	Reset
3:0	trRamVerMajor	Trace RAM Sink Component Major Version. Value 1 means the component is compliant with this document.	RO	1
7:4	trRamVerMinor	Trace RAM Sink Component Minor Version. Value 0 means the component is compliant with this document.	RO	0
11:8	trRamCompType	Trace RAM Sink Component Type (RAM Sink)	RO	0x9
12	trRamHasSRAM	This RAM Sink supports SRAM mode	RO	SD
13	trRamHasSMEM	This RAM Sink supports SMEM (System Memory) mode	RO	SD
23:14	—	Reserved for future versions of this standard	—	0
31:24	—	Reserved for vendor specific implementation details	—	SD



Single RAM Sink may support both SRAM and SMEM modes, but not both may be enabled at the same time. It is also possible to have more than one RAM Sink in a system.

Table 38. Register: **trRamStartLow**: Trace RAM Sink Start Register (**trBaseRamSink+0x010**)

Bit	Field	Description	RW	Reset
1:0	—	Always 0 (two LSB of 32-bit address)	RO	0
31:2	trRamStartLow	Byte address of start of trace sink circular buffer. It is always aligned on at least a 32-bit/4-byte boundary. An SRAM sink will usually have trRamStartLow fixed at 0.	WARL	Undef or fixed to 0

For a bus with an address larger than 32-bit, corresponding **High** registers define the MSB part of such a larger address.

Table 39. Register: **trRamStartHigh**: Trace RAM Sink Start High Bits Register (**trBaseRamSink+0x014**)

Bit	Field	Description	RW	Reset
31:0	trRamStartHigh	High order bits (63:32) of trRamStart register.	WARL	Undef

Table 40. Register: **trRamLimitLow**: Trace RAM Sink Limit Register (**trBaseRamSink+0x018**)

Bit	Field	Description	RW	Reset
1:0	—	Always 0 (two LSB of 32-bit address)	RO	0
31:2	trRamLimitLow	Highest absolute 32-bit part of address of trace circular buffer. The trRamWP register is reset to trRamStart after a trace word has been written to this address.	WARL	Undef

Table 41. Register: **trRamLimitHigh**: Trace RAM Sink Limit High Bits Register (**trBaseRamSink+0x01C**)

Bit	Field	Description	RW	Reset
31:0	trRamLimitHigh	High order bits (63:32) of trRamLimit register.	WARL	Undef

Table 42. Register: **trRamWPLow**: Trace RAM Sink Write Pointer Register (**trBaseRamSink+0x020**)

Bit	Field	Description	RW	Reset
0	trRamWrap	Set to 1 by hardware when trRamWP wraps. It is only set to 0 if trRamWPLow is written	WARL	Undef
1	—	Always 0 (bit B1 of 32-bit address)	RO	0
31:2	trRamWPLow	Absolute 32-bit part of address in trace sink memory where next trace message will be written. Fixed to a natural boundary. After a trace word write occurs while trRamWP = trRamLimit , trRamWP is set to trRamStart .	WARL	Undef

Table 43. Register: trRamWPHigh: Trace RAM Sink Write Pointer High Bits Register (trBaseRamSink+0x024)

Bit	Field	Description	RW	Reset
31:0	trRamWPHigh	High order bits (63:32) of trRamWP register.	WARL	Undef

Table 44. Register: trRamRPLow: Trace RAM Sink Read Pointer Register (trBaseRamSink+0x028)

Bit	Field	Description	RW	Reset
1:0	—	Always 0 (two LSB of 32-bit address)	RO	0
31:2	trRamRPLow	Absolute 32-bit part of address in trace circular memory buffer visible through trRamData. trRamRP auto-increments following an access to trRamData. After a trace word read occurs while trRamRP = trRamLimit, trRamRP is set to trRamStart. Required for SRAM mode and optional for SMEM mode.	WARL	Undef

Table 45. Register: trRamRPHigh: Trace RAM Sink Read Pointer High Bits Register (trBaseRamSink+0x02C)

Bit	Field	Description	RW	Reset
31:0	trRamRPHigh	High order bits (63:32) of trRamRP register.	WARL	Undef

Table 46. Register: trRamData: Trace RAM Sink Data Register (trBaseRamSink+0x040)

Bit	Field	Description	RW	Reset
31:0	trRamData	Read (and optional write) value for trace sink memory access. SRAM is always accessed by 32-bit words through this path regardless of the actual width of the sink memory. Required for SRAM mode and optional for SMEM mode.	R or RW	Undef



When trace capture was wrapped around (**trRamWrap** = 1) beginning of trace is not available and oldest packets/messages in the trace buffer (starting at address in **trRamWP**) will most likely not be complete. Trace decoders must look for the start of a message. Also when trace is stopped on wrap around, the very last message recorded in trace memory may not be complete.

The table below shows typical Trace RAM Sink configurations. Implementing other configurations is not suggested as trace tools may not support it without adjustments.

Table 47. Typical Trace RAM Sink Configurations

Mode	trRamStart	trRamLimit	trRamWP	trRamRP	trRamData
SRAM	0	Hard coded to max size ($2^M - A$) at reset, but can be possibly trimmed	Required	Required	Required
SMEM Generic	Any (2^N aligned)	Any (trRamStart + $2^M - A$) - must be set by trace tool	Required	Not implemented	Not implemented
SMEM Fixed	Fixed (2^N aligned)	Fixed to max size at reset (trRamStart + $2^M - A$), but can be possibly trimmed	Required	Not implemented	Not implemented



Value **A** means alignment which depends on memory access width. If we have memory access width of 32-bits, $A=4$ and value of **trRamLimit** register should be 0x...FC. Some implementations may impose bigger alignment of trace data (to allow more efficient transfer rates) for SMEM mode. For SRAM mode **A** must be 4 as access to trace via **trRamData** is always 32-bits wide.

7.1. Accessing and Detecting RAM Sink Registers

Trace tool should start interacting with Trace RAM Sink by releasing RAM Sink from reset by setting `trRamActive = 1` and waiting for this bit to be set. After that it should verify `trRamEmpty = 1`, read `trRamImpl` and verify `trRamCompType` and `trRamVer??` fields. Values of `trRamHasSRAM/SMEM` fields will provide main types of RAM Sink being implemented.

Later `trRamMode` should be set (depending on desired RAM Sink mode). It is important to set this field first as other registers may behave differently for SRAM and SMEM modes.

In SRAM mode, the trace memory is dedicated for trace storage and `trRamStart??` registers should not be settable (usually both not implemented and return 0). `trRamLimitLow` register may be either hardcoded (to reflect physical SRAM size) or writable (allowing trimming RAM size allowing faster wrap-around or sharing the same memory with some other components in the system). The `trRamLimitHigh` register should not be implemented as it is not practical to have more than 4GB of dedicated on-chip RAM storage.

Detection of valid ranges of each `trRamStart??` and `trRamLimit??` registers should be performed by writing 0 and 0xFFFFFFFF. After setting 0, the lowest possible value must be set. After setting 0xFFFFFFFF the highest possible value must be set. If the highest value for `trRamStartHigh` or `trRamLimitHigh` is 0, it means the register is NOT implemented.

Some implementations may provide different limits for different start addresses, so the trace tool should always set `trRamStart??` registers first - this option can be used when a particular implementation has two different RAM regions (each with different physical memory size).

Not every value may be settable in `trRamStart/Limit` registers. Value written may be trimmed (for example aligned on a particular 2^N boundary) and a trace tool should verify values being written. In case accepted values are different from what was provided by the user, a message should be printed which may allow the user to adjust (possibly suboptimal) settings.

Registers `trRamStart??` and `trRamLimit??` are usually set at the beginning of a debug/trace session and never changed.



In SMEM mode (`trRamMode = 1`) trace tool should never set `trRamStart??` and `trRamLimit??` registers outside of range provided by the user as otherwise raw trace being written to memory may corrupt running code and/or data or stack. This type of errors may be very difficult to diagnose as in complex system code (or data) being overwritten by trace may be used way, way later after actual corruption was made.

Having both `trRamStart/Limit??` registers set, the tool should try to set `trRamRP??` to the same value as `trRamLimit??`. If it is settable, it means that the `trRamData` register should be used to read the trace. Otherwise collected trace must be read using normal, physical memory accesses (in range defined by `trRamStart/Limit??` registers).

Before enabling RAM Trace Sink (by setting `trRamEnable = 1`) the trace tool should set `trRamWP??` registers (usually to the same values as in `trRamStart??` register). Enabling trace must NOT change any of `trRamStart/Size/WP/RP??` registers. Just after the trace is enabled `trRamWP??` may change because of trace being added to trace memory.

After trace is enabled and active (`trRamEnable = 1` or `trRamEmpty = 0`), the trace tool should NOT write any of `trRamStart/Limit/WP??` registers.

Setting `trRamRP` and reading `trRamData` may be attempted while trace is active, but support for reading

SRAM trace while trace is active may not always be implemented. In such a case write to **trRamRP** must be ignored and **trRamData** read must not advance **trRamRP**. Reading the trace in the SMEM mode via normal memory reads is always allowed.



*Even if reading trace (while trace is active) is implemented, circular trace buffer may be overwritten even several times, so values being read by **trRamData** will be of no use. However, when trace is started/stopped by infrequent triggers, reading SRAM trace may be useful. Also, the very last packet in memory may be incomplete as the last trace word may be buffered inside (and **trRamEmpty** = 0 will be observed).*



*Trace RAM Sink may implement writing trace by writing to **trRamData**, but this mode is usable only for testing, so will most likely not be implemented. Trace tool is not required to support writing to the **trRamData** register.*

Chapter 8. Trace Funnel

The Trace Funnel combines messages/packets from multiple sources into a single trace stream. It is implementation dependent how many incoming messages/packets are accepted before it is switching to another input source and in what order. But a continuous stream of messages/packets at one input cannot cause other inputs to not be handled. Suggested implementation would be to process just a single message/packet from each input in a round-robin fashion.

Table 48. Register: trFunnelControl: Trace Funnel Control Register (trBaseFunnel+0x000)

Bit	Field	Description	RW	Reset
0	trFunnelActive	Primary activate/reset bit for trace funnel. When 0, the Trace Funnel may have clocks gated off or be powered down, and other register locations may be inaccessible. Hardware may take an arbitrarily long time to process power-up and power-down and will indicate completion when the read value of this bit matches what was written. See Reset and Discovery chapter for more details.	RW	0
1	trFunnelEnable	1: Trace Funnel enabled. Setting trFunnelEnable to 0 flushes any queued trace data to output. See Enabling and Disabling chapter for more details.	RW	0
2	—	Reserved	—	0
3	trFunnelEmpty	Reads 1 when Trace Funnel internal buffers are empty	RO	1
31:4	—	Reserved	—	0

Table 49. Register: trFunnelImpl: Trace Funnel Implementation Register (trBaseFunnel+0x004)

Bit	Field	Description	RW	Reset
3:0	trFunnelVerMajor	Trace Funnel Component Major Version. Value 1 means the component is compliant with this document.	RO	1
7:4	trFunnelVerMinor	Trace Funnel Component Minor Version. Value 0 means the component is compliant with this document.	RO	0
11:8	trFunnelCompType	Trace Funnel Component Type (Trace Funnel)	RO	0x8
23:12	—	Reserved for future versions of this standard	—	0
31:24	—	Reserved for vendor specific implementation details	—	SD

Table 50. Register: trFunnelDisInput: Disable Individual Funnel Inputs (trBaseFunnel+0x008)

Bit	Field	Description	RW	Reset
15:0	trFunnelDisInput	1: Funnel input #n (bit position in register) is disabled. Incoming messages are read from disabled input but discarded.	WARL	Undef
31:16	—	Reserved	—	0



trFunnelDisInput register is optional. When not implemented (or never set) it will read as 0, which means that all inputs are always enabled. When implemented, it can be set to 0xFFFF to detect which inputs may be disabled in that trace funnel. Disabling inputs is needed when a single trace encoder may provide output to more than one possible active destination/sink. This register can be also used by trace tools to easily configure a trace in complex systems. Without the ability to disable individual funnel inputs, the trace tool must assure all trace sources which should not be traced are disabled.

8.1. Timestamp Unit

Trace Funnel may optionally include Timestamp Unit. It is described inside of the Trace Encoder chapter above.

Chapter 9. Trace PIB Sink

Trace data may be sent to chip pins through an interface called the Pin Interface Block (PIB). This interface typically operates at a few hundred MHz and can sometimes be higher with careful constraints and board layout or by using LVDS or other high-speed signal protocol. PIB may consist of just one signal and in this configuration may be called SWT (Serial-Wire Trace). Alternative configurations include a trace clock (TRC_CLK) and 1/2/4/8/16 parallel trace data signals (TRC_DATA) timed to that trace clock. WARL register fields are used to determine specific PIB capabilities.

The modes and behavior described here are intended to be compatible with trace probes available in the market.

PIB Register Interface

Table 51. Register: trPibControl: PIB Sink Control Register (trBasePib+0x000)

Bit	Field	Description	RW	Reset
0	trPibActive	Primary activate/reset bit for PIB Sink component. When 0, the PIB Sink may have clocks gated off or be powered down, and other register locations may be inaccessible. Hardware may take an arbitrarily long time to process power-up and power-down and will indicate completion when the read value of this bit matches what was written. See Reset and Discovery chapter for more details.	RW	0
1	trPibEnable	0: PIB does not accept input but holds output(s) at idle state defined by pibMode. 1: Enable PIB to generate output. See Enabling and Disabling chapter for more details.	RW	0
2	—	Reserved	—	0
3	trPibEmpty	Reads 1 when PIB internal buffers are empty.	RO	1
7:4	trPibMode	Select mode for output pins. Allowed values are described in the Allowed PIB Configurations table below.	WARL	Undef
8	trPibClkCenter	In parallel modes, adjust TRC_CLK timing to the center of the bit period. This can be set only if trPibMode selects one of the parallel protocols.	WARL	Undef
9	trPibCalibrate	Set this to 1 to generate a repeating calibration pattern to help tune a probe's signal delays, bit rate, etc. In this mode input to the sink is not consumed. The calibration pattern is described below.	WARL	Undef
11:10	—	Reserved	—	0
14:12	trPibAsyncFreq	0: Alignment synchronization (Async) packets disabled (may be the only choice for some protocols) 1-7: Different levels of alignment synchronization (bigger number, bigger distance). Details should be defined in the specification of each trace protocol.	WARL	Undef
15	—	Reserved	—	0
31:16	trPibDivider	Timebase selection for the PIB module. The input clock is divided by trPibDivider + 1. PIB data is sent at either this divided rate or 1/2 of this rate, depending on trPibMode. Width is implementation dependent. After the PIB reset value of this field should be set to safe (not too fast clock) setting for a particular SoC. Trace tools may set smaller values to utilize higher bandwidth.	WARL	Undef

Table 52. Register: **trPibImpl**: Trace PIB Implementation Register (**trBasePib+0x004**)

Bit	Field	Description	RW	Reset
3:0	trPibVerMajor	Trace PIB Sink Component Major Version. Value 1 means the component is compliant with this document.	RO	1
7:4	trPibVerMinor	Trace PIB Sink Component Minor Version. Value 0 means the component is compliant with this document.	RO	0
11:8	trPibCompType	Trace PIB Sink Component Type (PIB Sink)	RO	0xA
23:12	—	Reserved for future versions of this standard	—	0
31:24	—	Reserved for vendor specific implementation details	—	SD

Software can determine what modes are available by attempting to write each mode setting to the WARL field **trPibMode** and reading back to see if the value was accepted.

Table 53. Allowed PIB Configurations

Mode	trPibMode	trPibClkCenter	Bit rate
Off	0	X	—
SWT Manchester	4	X	1/2
SWT UART	5	X	1
TRC_CLK + 1 TRC_DATA	8	0	1
TRC_CLK + 2 TRC_DATA	9	0	1
TRC_CLK + 4 TRC_DATA	10	0	1
TRC_CLK + 8 TRC_DATA	11	0	1
TRC_CLK + 16 TRC_DATA	12	0	1
TRC_CLK + 1 TRC_DATA	8	1	1/2
TRC_CLK + 2 TRC_DATA	9	1	1/2
TRC_CLK + 4 TRC_DATA	10	1	1/2
TRC_CLK + 8 TRC_DATA	11	1	1/2
TRC_CLK + 16 TRC_DATA	12	1	1/2

Since the PIB supports many different modes, it is necessary to follow a particular programming sequence:

- Activate the PIB by setting **trPibActive**.
- Set the **trPibMode**, **trPibDivider**, **trPibClkCenter**, and **trPibCalibrate** fields. This will set the TRC_DATA outputs to the quiescent state (whether that is high or low depends on **trPibMode**) and start TRC_CLK running.
- Activate the receiving device, such as a trace probe. Allow time for PLL to sync up, if using a PLL with a parallel PIB mode.
- Set **trPibEnable**. This enables the PIB to generate output either immediately (calibration mode) or when the Trace Encoder or Trace Funnel begins sending trace messages/packets.

9.1. Order of bits and bytes

- Trace messages/packets are considered as sequences of bytes and are always transmitted with least significant bits/bytes first.
- In 16-bit mode (**trPibMode** == 12) the byte transmitted on bits #0-#7 is considered first and most

significant bits#8-#15 are transmitting second byte.

- Idle sequences (no message/packet to be sent) are transmitted between messages.
 - Idle sequence depends on trace protocol and must allow detection of the start of first byte of message/packet following the idle sequence.
 - Idle sequences may be different and should be defined by trace protocols.

9.2. PIB Parallel Protocol

Traditionally, off-chip trace has used this protocol. There are several parallel data signals (TRC_DATA0..15) and one continuously-running trace clock (TRC_CLK). The data rate of parallel signals can be much higher than either of the serial-wire protocols.

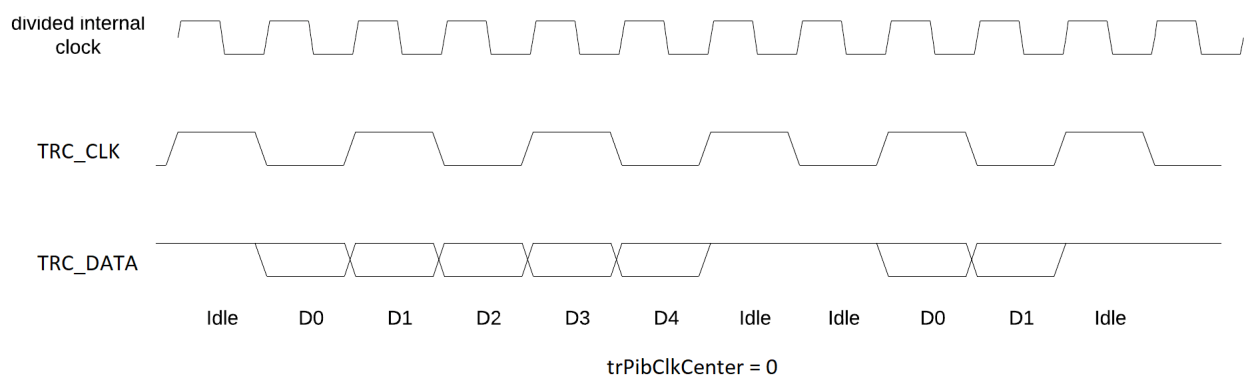
This protocol is oriented to send full, variable length trace messages/packets rather than fixed-width trace words.

When a message start is detected, this sample and possibly the next few (depending on the width of TRC_DATA) are collected until a complete byte has been received. Bytes are transmitted least significant bit first, with TRC_DATA[0] representing the least significant bit in each beat of data. The receiver continues collecting bytes until a complete message has been received. The criteria for this depends on the trace format. After the last byte of a message, the data signals may then go to their idle state or a new message may begin in the next trace clock edge.

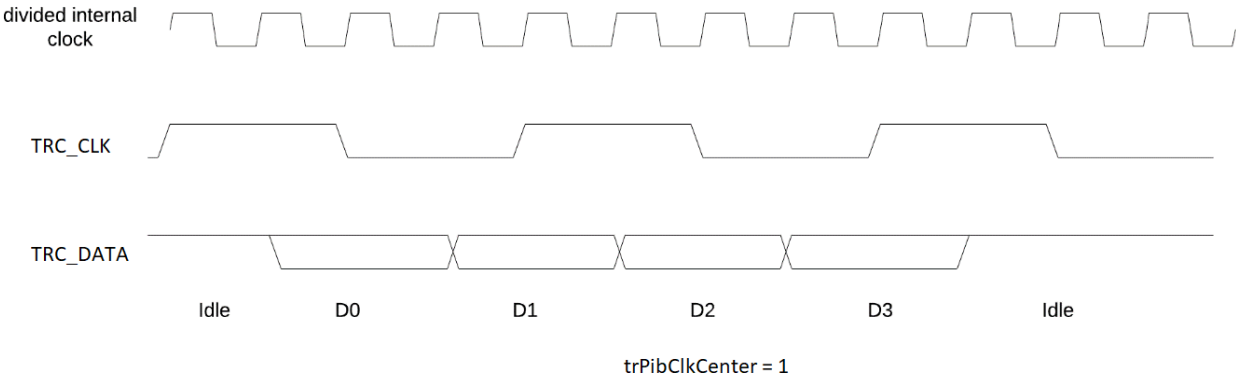
9.2.1. PIB Clock Center

The trace clock, TRC_CLK, normally has edges coincident with the TRC_DATA edges. Typically, a trace probe will delay trace data or use a PLL to recover a sampling clock that is twice the frequency of TRC_CLK and shifted 90 degrees so that its rising edges occur near the center of each bit period. If the PIB implementation supports it, the debugger can set `trPibClkCenter` to change the timing of TRC_CLK so that there is a TRC_CLK edge at the center of each bit period on TRC_DATA. Note that this option cuts the data rate in half relative to normal parallel mode and still requires the probe to sample TRC_DATA on both edges of TRC_CLK.

This example shows 8-bit parallel mode with `trPibClkCenter = 0` transmitting a 5-byte message/packet followed by a 2-byte message/packet.



And an example showing 8-bit parallel mode transmitting a 4-byte packet with `trPibClkCenter = 1`

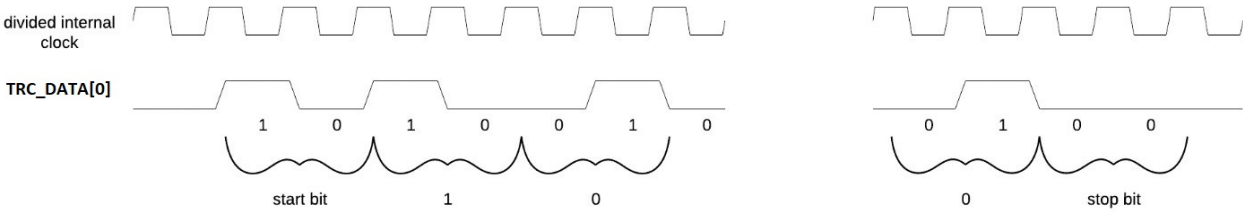


9.3. SWT Manchester Protocol

In this mode, the PIB outputs complete trace messages encapsulated between a start bit and a stop bit. Each bit period is divided into 2 phases and the sequential values of the TRC_DATA[0] pin during those 2 phases denote the bit value. Bits of the message are transmitted LSB first. The idle state of TRC_DATA[0] is low in this mode.

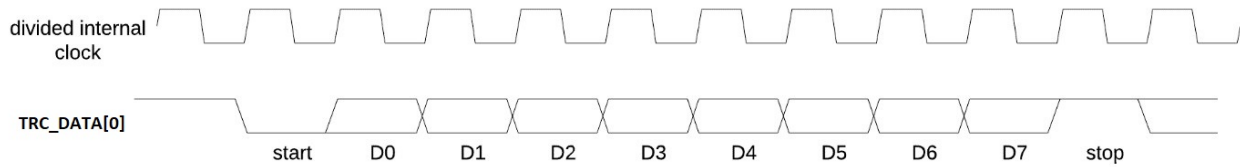
Table 54. Manchester Encoding Patterns

Bit	Phase 1	Phase 2
start	1	0
logic 0	0	1
logic 1	1	0
stop/idle	0	0



9.4. SWT UART Protocol

In UART protocol, the PIB outputs bytes of a trace message encapsulated in a 10-bit packet consisting of a low start bit, 8 data bits, LSB first, and a high stop bit. Another packet may begin immediately following the stop bit or there may be an idle period between packets. When no data is being sent, TRC_DATA[0] is high in this mode.



9.5. Calibration Mode

In optional calibration mode, the PIB transmits a repeating pattern. Probes can use this to automatically tune input delays due to skew on different PIB signal lines and to adjust to the transmitter's data rate (`trPibDivider` and `trPibClkCenter`). Calibration patterns for each mode are listed below.

Table 55. PIB Calibration Patterns

Mode	Calibration Bytes	Wire Sequence
UART, Manchester	AA 55 00 FF	alternating 1/0, then all 0, then all 1
1-bit parallel	AA 55 00 FF	alternating 1/0, then all 0, then all 1
2-bit parallel	66 66 CC 33	2, 1, 2, 1, 2, 1, 2, 1, 0, 3, 0, 3, 0, 3, 0
4-bit parallel	5A 5A FO OF	A, 5, A, 5, 0, F, F, 0
8-bit parallel	AA 55 00 FF	AA, 55, 00, FF
16-bit parallel	AA AA 55 55 00 00 FF FF	AAAA, 5555, 0000, FFFF



Calibration mode may be used even by probes which do not support calibration of trace just to assure trace routing on PCB is correct and PIB is correctly enabled. It may be also possible to use calibration mode to check trace signal routing from SoC using scope or logic analyzer.

Chapter 10. Trace ATB Bridge

Some SoCs may have an Advanced Trace Bus (ATB) infrastructure to manage trace produced by other components. In such systems, it may be desired to route entire RISC-V trace stream to the ATB through an ATB Bridge. This module manages the interface to ATB, generating ATB trace records that encapsulate RISC-V trace produced by the Trace Encoder[s] and/or Trace Funnel[s]. There is a control register that includes trace on/off control and a field allowing software to set the ID to be used on the ATB bus. This ID allows software to extract entire RISC-V trace from the combined trace. This interface is compatible with AMBA 4 ATB v1.1.

Table 56. Register: trAtbBridgeControl: ATB Bridge Control Register (trAtbBridgeBase+0x000)

Bit	Field	Description	RW	Reset
0	trAtbBridgeActive	Primary activate/reset for the ATB Bridge. When 0, the ATB Bridge may have clocks gated off or be powered down, and other register locations may be inaccessible. Hardware may take an arbitrarily long time to process power-up and power-down and will indicate completion when the read value of this bit matches what was written. See Reset and Discovery chapter for more details.	RW	0
1	trAtbBridgeEnable	1: ATB Bridge enabled. Setting trAtbBridgeEnable to 0 flushes any queued trace data to ATB. See Enabling and Disabling chapter for more details.	RW	0
2	—	Reserved	—	0
3	trAtbBridgeEmpty	Reads 1 when ATB Bridge internal buffers are empty	RO	1
7:4	—	Reserved	—	0
14:8	trAtbBridgeID	ID of this node on ATB. Values of 0x00 and 0x70-0x7F are reserved by the ATB specification and should not be used.	RW	Undef
31:15	—	Reserved	—	0

Table 57. Register: trAtbBridgeImpl: ATB Bridge Implementation Register (trAtbBridgeBase+0x004)

Bit	Field	Description	RW	Reset
3:0	trAtbBridgeVerMajor	ATB Bridge Component Major Version. Value 1 means the component is compliant with this document.	RO	1
7:4	trAtbBridgeVerMinor	ATB Bridge Component Minor Version. Value 0 means the component is compliant with this document.	RO	0
11:8	trAtbBridgeCompType	ATB Bridge Component Type (ATB Bridge)	RO	0xE
14:12	trAtbBridgeAsyncFreq	0: Alignment synchronization (Async) packets disabled (may be the only choice for some protocols) 1-7: Different levels of alignment synchronization (bigger number, bigger distance). Details should be defined the specification of each trace protocol.	WARL	Undef
23:15	—	Reserved for future versions of this standard	—	0
31:24	—	Reserved for vendor specific implementation details	—	SD

An implementation determines the data widths of the connection from the Trace Encoder or Trace Funnel and of the ATB port.

ATB Bridge may optionally insert ATB alignment synchronization packets (controlled by **trAtbBridgeAsyncFreq** field) allowing trace decoding software to detect ATB packet boundaries. Not all protocols may require it.

Chapter 11. Additional Material

11.1. Minimal Implementation

This (non-normative) chapter gives an of what needs to be done to put together complete RISC-V trace implementation without getting familiar with every detail of every register.

Minimal General Registers/Fields

These requirements are applicable to the entire trace sub-system.

- One **Trace Encoder** per hart being traced is required.
- At least one of **Trace RAM** or **Trace PIB** sinks or **Trace ATB Bridge** is required as the final destination of an encoded RISC-V trace.
 - Implementations providing custom transport only are NOT considered fully compliant with this specification as custom trace tools will be needed.
- Each trace component in a system is required to implement **tr??Control** and **tr??Impl** registers.
 - **tr??Active** bit must be settable to 0 or 1, although reset itself is NOT required.
 - **tr??Enable** bit must be settable to 0 or 1 and must support flushing (if applicable) when changed from 1 to 0.
 - **tr??Empty** bit must read as 0 when the trace component has some trace data internally buffered (if trace component it not buffering any trace data, this bit may be hard coded to 1).
 - **tr??VerMajor**, **tr??VerMinor** and **tr??CompType** must be implemented.

Minimal Trace Encoder Register/Fields

- Bit **trTeInstTracing** must be implemented (to start/stop instruction trace output from Trace Encoder).
- One of **trTeInstMode** = 3 (Baseline instruction trace) or 6 (Optimized instruction trace) must be implemented (can be a hard coded value).
- At least one of the non-0 values of **trTeInstSyncMode** must be settable (or hard coded).
- Field **trTeFormat** must correspond to implemented trace protocol (0 for E-Trace or 1 for N-Trace).
- Fields **trTeProtocolMajor** and **trTeProtocolMinor** must return versions of implemented protocol.
- All other registers/fields/bits may be tied to 0.

Minimal Trace RAM Sink Register/Fields

SRAM mode only:

- Bit **trRamHasSRAM** must be tied to 1 and **trRamMode** must be tied to 0.
- Bit **trRamWrap** must be implemented.
- Register **trRamLimitLow** must be implemented but can be hard coded to value '2^M-4' (address 0x..FC).
- Register **trRamWPLow** must at least accept a write of 0.
- Register **trRamRPLow** must accept any 32-bit aligned value in inclusive range < 0 .. **trRamLimitLow** >.
 - If width of access to SRAM is wider than 32-bits any 32-bit aligned value of **trRamRP** must be

allowed and reads must be buffered.

- Register **trRamData** may be implemented for reading only.
- All other registers/fields/bits may be tied to 0.

SMEM mode only:

- Bit **trRamHasSMEM** must be tied to 1 and **trRamMode** must be tied to 1.
- Bit **trRamWrap** must be implemented.
- Register **trRamStart** must be implemented but can be hard coded to value '2^N' (address 0x..00).
- Register **trRamLimit** must be implemented but can be hard coded to value '2^N + 2^M-4' (address 0x..FC).
- Registers **trRamWP** must accept any 32-bit aligned value in inclusive range $< \text{trRamStart} .. \text{trRamLimit}$.
- All other registers/fields/bits may be tied to 0.

Minimal Trace PIB Sink Register/Fields

It is hard to define required mode as it depends on SoC bandwidth requirements and capabilities, but some general guidance may be provided.

- 4-bit mode is supported by most (if not all) trace probes and less expensive MIPI20 connectors can be used.
 - 1-bit and 2-bit modes should be only used when there are critical constraints on the number of MCU pins. Not all trace probes may support these modes.
- Serial mode should be only considered when either limited trace is required, or cores run slowly. Not all trace probes may support this mode and max allowed speeds may vary.
 - Manchester encoding is self-synchronizing and may provide a more reliable trace. However UART mode may provide better bandwidth. It is suggested to support both.
- 8-bit and 16-bit modes will provide better bandwidth, but require a more expensive Mictor connector and only more advanced trace probe models may support it.
- It is suggested to provide as fast as possible trace logic clock, and allow a trace tool to set the divider in the **trPibDivider** field.
- For TRC_CLK frequencies higher than 50MHz, it is suggested to provide a calibration mode.
 - If possible, implement **trPibClkCenter** for better flexibility.

Minimal ATB Bridge Register/Fields

- Field **trAtbBridgeID** must be settable by trace tool (hard coded ID may not be handled by all trace tools).

11.2. Reset and Discovery

This chapter describes what trace tools should do to reset and discover trace features.



Trace tools must be provided with base addresses of all trace components.

Only the **tr??Active** and the **tr??Enable** bits are reset to 0 on power-up.

These **tr??Active** bits act as independent resets for the respected trace components:

- **trTeActive** - reset for Trace Encoder component (this will disable encoder from single hart)
- **trFunnelActive** - reset for Trace Funnel component
- **trPibActive** - reset for PIB component (resets Pin Interface Block only)
- **trRamActive** - reset for RAM component (resets RAM Sink only)
- **trAtbBridgeActive** - resets ATB Bridge component (resets ATB Bridge interface)
- **trTsActive** - resets the Timestamp Unit sub-component (resets timestamp generation logic)

When component is held in reset (**tr??Active** is 0), then **tr??Enable** bit must be reset to 0 as well (what makes component disabled).

Releasing components from reset (by setting **tr??Active** to 1) may take time - debug tools should monitor (with reasonable timeout) if the appropriate bit changed from 0 to 1.

As not all fields/registers are affected by reset (defined as **Undef**), the trace tools must initialize (usually, a write of a value 0) several registers to assure that trace component is in a predictable state.



- *Some of the reset values are defined as **SD** (system dependent) and these values should reset as well and each time to the same value as would be after power-up.*
- *Most of fields/registers have **Undef** specified in reset behavior of the field. It should not prevent some implementations from resetting these.*

Reset and Discovery should be performed as follows:

- Reset the component by setting **tr??Active** = 0.
 - This should be done by writing a value 0x0 to **tr??Control** register.
- Read-back and wait until **tr??Active** = 0 is read, which means that a component reached a reset state.
- Release a component from reset by setting **tr??Active** = 1.
 - This should be done by writing a value 0x1 to **tr??Control** register. This write will reset most of other enable/mode bits in this register and all WARL and read-only fields will be set to defaults.
- Read-back and wait until **tr??Active** = 1 is read, which means that a component was released from reset.
 - In this moment **tr??Enable** is set to 0 and the component is not yet enabled.
 - Component clock should be enabled to allow programming of other registers.
- Optionally save **tr??Control** register as it holds all reset values of all fields. It may be cached/shadowed, and trace tool may execute faster write-only (instead a read-modify-write) operations.

- Handle `tr??VerMinor/Major` as described in 'Versioning of Components' chapter.
 - If `tr??VerMajor` is 0 (for Trace Encoder component) either handle it as pre-ratified/initial version 0 or generate fatal error with an appropriate error message.
- Read `tr??Impl` and compare `tr??ComType` field with expected value.
- Set some WARL fields and read back to discover supported component configuration - make sure the component is NOT enabled (by setting `tr??Enable` to 1) by mistake.
- Configure some initial values in all needed registers/fields. Optionally Read-back to assure these are set properly.

The table below is showing what registers needs to be written to have each trace component in identical, predictable state.

Table 58. Trace Components Reset

Component	Register	Write	Notes
Trace Encoder	<code>trTeControl</code>	0x1	Release from reset and set all defaults.
	<code>trTeInstFeatures</code>	0x0	Force minimal sub-set of features.
	<code>trTeInstFilters</code>	0x0	Disable all filters.
	<code>trTeDataControl</code>	0x0	Disable data trace.
	<code>trTeDataFilters</code>	0x0	Disable filters for data trace.
	<code>trTeTrigDbgControl</code>	0x0	Disable all triggers from Debug module.
	<code>trTeTrigExtInControl</code>	0x0	Disable all external trigger inputs.
	<code>trTeTrigExtOutControl</code>	0x0	Do not general any external trigger outputs.
	<code>trTsControl</code>	0x0	Keep timestamp unit in reset.
Trace Funnel	<code>trFunnelControl</code>	0x1	Release from reset and set all defaults.
	<code>trFunnelDisInput</code>	0x0	Make sure all inputs are enabled.
	<code>trTsControl</code>	0x0	Keep timestamp unit in reset.
Trace RAM Sink	<code>trRamControl</code>	0x1	Release from reset and set all defaults.
Trace PIB Sink	<code>trPibControl</code>	0x1	Release from reset and set all defaults.
Trace ATB Bridge	<code>trAtbBridgeControl</code>	0x1	Release from reset and set all defaults.

As we are dealing with several independent components, it is important to assure that the component which is in reset (or powered down) is keeping its outputs on safe values, so garbage trace data is not emitted.

In general, it is safer to power-up and enable components starting from sinks/bridges, followed by Trace Funnels and Trace Encoders as last. Each implementation should test this sequence to assure trace tools are working seamlessly.



Pre-release version of this specification specified that most of fields and registers were reset. It was suggested (by Architecture Review Committee) that reset logic is made minimal to follow RISC-V ISA style. Older implementations (which reset more bits) are still compatible with a ratified version. These implementations do not have to change as it is perfectly OK to reset more fields and registers.

11.3. Enabling and Disabling

Enabling should work as follows:

- Release all needed components from reset by setting `tr??Active = 1` as described above.
- Set desired mode and verify if that mode is set (regardless of discovery results).
- For RAM Sink:
 - Setup needed addresses (if possible and desired) as these may not reset.
- For PIB Sink:
 - Calibrate PIB (if possible and desired).
 - Start physical trace capture (trace probe dependent).
- Configure RAM Sink/PIB Sink/ATB Bridge in appropriate mode.
 - Verify if a particular mode is set.
- Set main enable for RAM Sink/PIB Sink/ATB Bridge component by setting `tr??Enable = 1`.
 - Read back and wait for confirmation (`tr??Enable = 1`).
- Enable Trace Funnel[s] in the same way.
- Configure and Enable Trace Encoder[s] in the same way (last should be writing `trTeEnable = 1` followed by reading to verify that it is set).
 - The `trTeInstMode` WARL field should be set to 6 - it may revert to different mode.
- Either manually set `trTeInstTracing=1` and/or `trTeDataTracing=1` bits or set triggers to start the trace.
- Start hart[s] to be traced (hart could be already running as well - in this case trace will be generated in the moment when `trTeInstTracing` or `trTeDataTracing` bit is set).
- Periodically read `trTeControl` for status of trace (as it may stop by itself due to triggers).
 - If RAM Sink was configured with `trRamStopOnWrap = 1`, read `trRamEnable` to see if RAM capture was stopped.



Discovery may not be necessary to enable and test the trace during development of SoC. However, a discovery must be possible and should be tested by SoC designer - this is necessary for trace tools to work with that SoC without any customization.



Trace tools may verify a particular setting once per session, so subsequent starts of trace may be faster.



Trace tools should provide configuration settings allowing more verbose logging mode during discovery and initialization, so potential compatibility issues may be solved.

Disabling the trace should work as follows:

- It is essential to disable the trace from encoders associated with stopped harts as entering debug mode is NOT flushing any trace pipelines.
- Disable and flush trace starting from Trace Encoders, then Trace Funnels and finally Trace Sinks or Trace Bridges.
 - Set `tr??Enable = 0` and wait for `tr??Enable = 0` and `tr??Empty = 1` for each trace component.

- It is important to do it in that order as otherwise data may be lost.
- Stop physical capture if PIB sink was enabled (trace probe dependent).
- Read the trace.
 - For RAM Trace Sink read `trRamWP` - depending on `trRamWrap` bit, you may read trace from two ranges.
 - For RAM Trace Sink in SRAM mode, set `trRamRP` and read `trRamData` multiple times.
 - For RAM Trace Sink in SMEM mode, read trace from system memory using memory read.
 - For PIB Trace Sink read trace from trace probe.
 - For ATB Bridge, read trace using Coresight components (ETB/TMC/TPIU).

Decoding trace

- Decoder (in most cases) must have access to code which is running on device either by reading it from device or from a file containing the code (binary/hex/srec/ELF).
- The trace collected by trace probes can be read and decoded while a trace is being captured (this is called trace streaming mode).
 - There is no guarantee that the last trace packet is completed until the trace is properly flushed and disabled.
- Decoding of the trace should never affect code being traced.

11.4. Pre-ratified/Initial Interface Version

The value of `trTeVerMajor` as 0 means this is the pre-ratified/initial version of this trace control interface.

Initially this specification was kept highly compatible, but after the decision to split all components into 4K regions it was very hard to track and list all changes and appropriate chapter was removed.

The migration path from 'ver 0' (for both IP providers and tool vendors) should not be hard as the main concepts remain unchanged.

Original donation from SiFive (which describes implementation of pre-ratified/initial version 0) can be found here: [RISC-V-Trace-Control-Interface-Proposed-20200612.pdf](#)



Not all trace tools may support pre-ratified/initial version 0. But all such tools should reject a version 0 with a very clear message.